Linear Algebra 2. Harbour.Space University

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November, 10th – November, 28th

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0 Introduction

Machine learning is the most successful
marketing campaign for rebranding linear algebra.

0.1 What we will not do

Create AGI. (Probably not.)

0.2 What we will do

Recap Linear Algebra 1. (I know that feel bro.)

Multiply matrices. Many matrices. By hand and on a computer. And then learn to see the geometric and physical meaning behind matrices and avoid unnecessary multiplications.

0.3 Grading criteria

30 % — In-class activities.
40 % — Graded home assignments.
30 % — In-class exams (midterm and final exams).

A bonus (1 % of the total grade) will be given to those who bring truly funny and witty linear algebra memes that I haven't seen yet. (The degree of wit will be determined by my subjective sense of humor.)

1 Recap: vectors, matrices, determinants

In this section we will recap the basic notions of linear algebra from Linear Algebra 1 that are used in this course without proofs.

Check your knowledge and see whether you can solve the following problems.

PROBLEM 1.1. Consider the vector space of polynomials with real coefficients and zero constant term

$$S = \{a_1 x + a_2 x^2 + \cdots + a_n x^n \mid a_i \in \mathbb{R}, n \in \mathbb{N}\}.$$

Find a basis and the dimension of this vector space.

PROBLEM 1.2. Consider a set of all continuous functions on the interval $[0, 1]$, such that $f(x) \neq 0$ for all $x \in [0, 1]$. Is this set a vector space? If yes, find a basis and the dimension of this vector space. If no, explain why.

PROBLEM 1.3. Let the vectors e_1, e_2, e_3 and x be given by their coordinates in some basis:

$$e_1 = (2, 1, -3), \quad e_2 = (3, 2, -5), \quad e_3 = (1, -1, 1), \quad x = (6, 2, -7).$$

Prove that (e_1, e_2, e_3) is also a basis of the space, and find the coordinates of x in this basis.

PROBLEM 1.4. Prove that each of the two given systems of vectors S and S' is a basis. Find the transition matrix C from S to S' ($e' = eC$):

$$S = ((1, 2, 1), (2, 3, 3), (3, 8, 2))$$
$$S' = ((3, 5, 8), (5, 14, 13), (1, 9, 2))$$

PROBLEM 1.5. Prove that the given system of vectors is linearly independent, and complete it to a basis of the space of rows:

$$a_1 = (2, 2, 7, -1), \quad a_2 = (3, -1, 2, 4), \quad a_3 = (1, 1, 3, 1).$$

PROBLEM 1.6. Find a basis and the dimension of the linear span of the following system of vectors:

$$a_1 = (1, 0, 0, -1), \quad a_2 = (2, 1, 1, 0), \quad a_3 = (1, 1, 1, 1), \quad a_4 = (1, 2, 3, 4), \quad a_5 = (0, 1, 2, 3).$$

PROBLEM 1.7. Given square matrices A and B of size 2025×2025 such that $\det(A) = 2$ and $\det(B) = 25$. Find $\det(BA)$.

PROBLEM 1.8. Given a square matrix A of size 2025×2025 with $\det(A) = 25$. Find $\det(A^{-1})$.

PROBLEM 1.9. Find the rank of the matrix:

$$A = \begin{pmatrix} 1 & 1 & 3 & 4 & 5 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 14 & 15 \\ 0 & 0 & 0 & 28 & 30 \end{pmatrix}.$$

PROBLEM 1.10. Find the dimension of the solution space of the homogeneous system $Ax = 0$ with matrix:

$$A = \begin{pmatrix} 1 & 1 & 0 & 4 & 5 \\ 0 & 2 & 5 & 2 & -1 \\ 0 & 0 & 0 & 14 & 15 \end{pmatrix}.$$

PROBLEM 1.11. Find the determinant of the following matrix:

$$\begin{pmatrix} 1 & 2 & 3 \\ 5 & 1 & 4 \\ 3 & 2 & 5 \end{pmatrix}.$$

PROBLEM 1.12. Let A be a square 2×2 matrix with $\det(A) = 2$, B be a square 5×5 matrix with $\det(B) = 5$, C be a rectangular 2×5 matrix and O be a zero matrix of size 5×2. Find

$$\det \begin{pmatrix} A & C \\ O & B \end{pmatrix}.$$

Problem 1.13. Find the determinant of the following 10×10 matrix:

$$\begin{pmatrix} 10 & 0 & 0 & \dots & 0 \\ 9 & 10 & 0 & \dots & 0 \\ 8 & 9 & 10 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \dots & 10 \end{pmatrix}.$$

Problem 1.14. Find the inverse of the following matrix:

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 10 \end{pmatrix}.$$

Problem 1.15. Does the set of all invertible 5×5 matrices with real entries form a vector space? If yes, find a basis and the dimension of this vector space. If no, explain why.

Problem 1.16. Does the set of all 5×5 matrices with real entries and zero trace ($a_{11} + a_{22} + a_{33} + a_{44} + a_{55} = 0$) form a vector space? If yes, find a basis and the dimension of this vector space. If no, explain why.

Problem 1.17. Let v be a vector with coordinates (v_1, v_2, v_3) in the basis e (where e consists of 3 vectors), and let C be the change-of-basis matrix ($e' = eC$). Find the coordinates of v in the basis e'.

2 Polynomials

2.1 Definition and basic properties

In this section we will consider polynomials with real coefficients. However, all the results (if not explicitly stated otherwise) will be valid for polynomials with complex coefficients and even for polynomials with coefficients from an arbitrary field (see the next section for definitions). That's why we formulate most results for the general case of polynomials with coefficients from a field F, but we will keep in mind $F = \mathbb{R}$ in this section.

Recall that a *polynomial* over F is a function of the form

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

where $a_n, a_{n-1}, \ldots, a_1, a_0 \in F$ and n is a non-negative integer. The set of all polynomials with coefficients from F is denoted by $F[x]$.

The *degree* of the polynomial is the highest power of the variable in the polynomial with a nonzero coefficient. We denote the degree of the polynomial $p(x)$ by $\deg p(x)$. The degree of the zero polynomial is defined to be $-\infty$ by convention.

Exercise 2.1. Prove that if $p(x), g(x)$ are nonzero polynomials, then $\deg(p(x)q(x)) = \deg p(x) + \deg q(x)$.

Exercise 2.2. Prove that if $p(x), g(x)$ are polynomials, then $\deg(p(x) + q(x)) \leqslant \max(\deg p(x), \deg q(x))$. Provide an example of polynomials $p(x)$ and $q(x)$ such that the equality is strict.

We can divide polynomials with real coefficients (for example, by long division). In general, we can divide one polynomial by another with a remainder. Assume we have two polynomials $p(x)$ and $q(x)$ with real coefficients. We can find two polynomials $s(x)$ and $r(x)$ with real coefficients such that $p(x) = q(x)s(x)+r(x)$ and $\deg r(x) < \deg q(x)$. The polynomial $s(x)$ is called the *quotient*, and the polynomial $r(x)$ is called the *remainder*. Note that the quotient and the remainder are unique.

Exercise 2.3. Find the quotient and the remainder of the division of $p(x) = x^3 + 2x^2 - x + 1$ by $q(x) = x^2 + 1$.

Definition 2.1. A number $\lambda \in F$ is called a *root* of the polynomial $p(x)$ if $p(\lambda) = 0$.

Lemma 2.2. $\lambda \in F$ is a root of the polynomial $p(x)$ if and only if $p(x)$ is divisible by $x - \lambda$.

Proof. Divide $p(x)$ by $x - \lambda$ with a remainder:

$$p(x) = (x - \lambda)q(x) + r(x),$$

where $r(x)$ is a polynomial of degree less than $\deg(x - \lambda) = 1$. Since $\deg r(x) < 1$, we have $r(x) = r$ for some number $r \in F$. Substituting $x = \lambda$ into the equation, we get $p(\lambda) = (x - \lambda)q(\lambda) + r = 0 + r = r$. Therefore, $r = 0$ ($p(x) = (x - \lambda)q(x)$) if and only if λ is a root of $p(x)$. □

Exercise 2.4. Prove the little Bézout's theorem: for any polynomial $p(x)$ and a number $\mu \in F$, the value $p(\mu)$ is equal to the remainder of the division of $p(x)$ by $x - \mu$.

Definition 2.3. We say that the *multiplicity* of a root λ of a polynomial $p(x)$ equals m if $p(x)$ is divisible by $(x - \lambda)^m$ and is not divisible by $(x - \lambda)^{m+1}$.

Lemma 2.4. A polynomial $p(x)$ of positive degree n has at most n roots.

Proof. By induction on n. For $n = 1$, the polynomial $a_1 x + a_0$ ($a_1 \neq 0$) has exactly one root $x = -\frac{a_0}{a_1}$. Assume that the statement is true for some $n \geqslant 1$. Let $p(x)$ be a polynomial of degree $n + 1$. If $p(x)$ has no roots, then the statement is true. If $p(x)$ has a root λ, then $p(x) = (x - \lambda)q(x)$ for some polynomial $q(x)$ of degree n. By the induction hypothesis, $q(x)$ has at most n roots. Therefore, $p(x)$ has at most $n + 1$ roots. □

Corollary 2.5. Coefficients of a polynomial over $\mathbb{R}$ ($\mathbb{C}$, or any other infinite field) are uniquely determined by the polynomial.

Proof. Assume that some polynomial (as a function) has two different representations. Then subtracting these representations, we get a polynomial of positive degree (why?) with infinitely many roots — all real numbers, which contradicts the lemma. □

Polynomials with coefficients from F form a vector space over F. One basis of this vector space is the set of polynomials $1, x, x^2, \ldots, x^n, \ldots$. Polynomials of degree at most n form a subspace of this vector space, which is denoted by $F_n[x]$.

Exercise 2.5. Find a basis and the dimension of the vector space $\mathbb{R}_n[x]$. Prove explicitly the linear independence of the basis.

The fact that polynomials form a vector space gives us a nice proof of the possibility (and uniqueness) of dividing polynomials with a remainder. Specifically, if we want to divide a polynomial $p(x)$ of degree n by a polynomial $q(x)$ of degree m, then we choose as a basis in $F_n[x]$ the set $1, x, x^2, \ldots, x^{m-1}, q(x), q(x)x, q(x)x^2, \ldots, q(x)x^{n-m}$. (Why is this a basis?) Then $p(x)$ can be uniquely represented in the form

$$p(x) = a_0 \cdot 1 + a_1 x + a_2 x^2 + \ldots + a_{m-1}x^{m-1} + a_m q(x) + a_{m+1}q(x)x + a_{m+2}q(x)x^2 + \ldots + a_{m+(n-m)}q(x)x^{n-m},$$

where the first m terms are the remainder of the division of $p(x)$ by $q(x)$, and the last $n-m+1$ terms are the product of $q(x)$ and the quotient.

2.2 Greatest common divisor

Recall that the greatest common divisor of two polynomials $p(x)$ and $q(x)$ is the polynomial that divides both $p(x)$ and $q(x)$ and is divisible by any other polynomial that divides both $p(x)$ and $q(x)$. Equivalently, the greatest common divisor is the polynomial of the highest degree that divides both $p(x)$ and $q(x)$. Notation: $\gcd(p(x), q(x))$, or just $(p(x), q(x))$.

The greatest common divisor of two polynomials can be found using the Euclidean algorithm. It is the last nonzero remainder in this algorithm. Moreover, by the extended Euclidean algorithm (i.e., by back-substitution in the Euclidean algorithm) we can find polynomials $s(x)$ and $t(x)$ such that $\gcd(p(x), q(x)) = s(x)p(x) + t(x)q(x)$.

Exercise 2.6. Find the greatest common divisor of the polynomials $p(x) = x^4 + 2x^3 - x^2 - 4x - 2$ and $q(x) = x^4 + x^3 - x^2 - 2x - 2$. Also represent it in the form $s(x)p(x) + t(x)q(x)$.

Exercise 2.7. Find all common zeros of polynomials $p(x) = 48x^3 - 22x^2 - 21x + 10$ and $q(x) = 32x^3 + 12x^2 - 12x - 5$. (Hint: first find the greatest common divisor.)

Definition 2.6. Two polynomials $p(x)$ and $q(x)$ are called coprime if $\gcd(p(x), q(x)) = 1$.

Analogously to the case of two polynomials, we define the greatest common divisor of several polynomials as a common divisor that is divisible by every common divisor of these polynomials.

Lemma 2.7. Let $p_1, p_2, \ldots, p_k$ be polynomials. Then there exist polynomials $s_1, s_2, \ldots, s_k$ such that

$$\gcd(p_1, p_2, \ldots, p_k) = s_1 p_1 + s_2 p_2 + \ldots + s_k p_k.$$

Proof. Prove by induction on k. (For $k = 1$ the statement is trivial.) For $k = 2$, the statement is true by the Euclidean algorithm. Assume that the statement is true for some $k \geqslant 2$. Let $p_1, p_2, \ldots, p_k, p_{k+1}$ be polynomials and

$$\gcd(p_1, p_2, \ldots, p_k) = s_1 p_1 + s_2 p_2 + \ldots + s_k p_k.$$

Since $\gcd(p_1, p_2, \ldots, p_k, p_{k+1}) = \gcd(\gcd(p_1, p_2, \ldots, p_k), p_{k+1})$, we can find two polynomials s_{k+1} and t such that

$$\gcd(\gcd(p_1, p_2, \ldots, p_k), p_{k+1}) = t \gcd(p_1, \ldots, p_k) + s_{k+1}p_{k+1},$$

i.e.

$$\gcd(p_1, p_2, \ldots, p_k, p_{k+1}) = ts_1 p_1 + ts_2 p_2 + \ldots + ts_k p_k + s_{k+1}p_{k+1}.$$

□

2.3 Problems

Problem 2.1. Find all roots of the following polynomials:

a) $p(x) = x^5 - 8x^4 + 16x^3 - 2x^2 - 17x + 10$;

b) $p(x) = x^4 - 15x^3 + 15x - 1$;

c) $p(x) = x^5 + 6x^4 + 24x^3 + 46x^2 + 39x + 12$.

PROBLEM 2.2. Divide the polynomial $f(x)$ by the polynomial $g(x)$:

a) $f(x) = 2x^4 - 3x^3 + 4x^2 - 5x + 6, \quad g(x) = x^2 - 3x + 1$;

b) $f(x) = x^3 - 3x^2 - x - 1, \quad g(x) = 3x^2 - 2x + 1$.

PROBLEM 2.3. Find the greatest common divisor of the following pairs of polynomials:

a) $x^4 + x^3 - 3x^2 - 4x - 1$ and $x^3 + x^2 - x - 1$;

b) $x^6 + 2x^4 - 4x^3 - 3x^2 + 8x - 5$ and $x^5 + x^2 - x + 1$;

c) $x^5 + 3x^2 - 2x + 2$ and $x^6 + x^5 + x^4 - 3x^2 + 2x - 6$;

d) $x^4 + x^3 - 4x + 5$ and $2x^3 - x^2 - 2x + 2$;

e) $x^5 + x^4 - x^3 - 2x - 1$ and $3x^4 + 2x^3 + x^2 + 2x - 2$;

f) $x^6 - 7x^4 + 8x^3 - 7x + 7$ and $3x^5 - 7x^3 + 3x^2 - 7$;

g) $x^5 - 2x^4 + x^3 + 7x^2 - 12x + 10$ and $3x^4 - 64x^3 + 5x^2 + 2x - 2$;

h) $x^5 + 3x^4 - 12x^3 - 52x^2 - 52x - 12$ and $x^4 + 3x^3 - 6x^2 - 22x - 12$;

i) $x^5 + x^4 - x^3 - 3x^2 - 3x - 1$ and $x^4 - 2x^3 - x^2 - 2x + 1$;

j) $x^4 - 4x^3 + 1$ and $x^3 - 3x^2 + 1$;

k) $x^4 - 10x^2 + 1$ and $x^4 - 4\sqrt{2}x^3 + 6x^2 + 4\sqrt{2}x + 1$.

PROBLEM 2.4. Find the greatest common divisor of the polynomials $f(x)$ and $g(x)$ and its linear expression through $f(x)$ and $g(x)$:

a) $f(x) = x^4 + 2x^3 - x^2 - 4x - 2, \quad g(x) = x^4 + x^3 - x^2 - 2x - 2$;

b) $f(x) = 3x^3 - 2x^2 + x + 2, \quad g(x) = x^2 - x + 1$.

PROBLEM 2.5. Find $(f(x), g(x), h(x))$ and represent it in the form $(f(x), g(x), h(x)) = s(x)f(x) + t(x)g(x) + u(x)h(x)$, where

a) $f(x) = 2x^3 - 3x^2 - 11x + 6, \quad g(x) = 2x^3 - x^2 - 50x + 25, \quad h(x) = x^3 - 4x^2 - 7x + 10$;

b) $f(x) = x^3 - 3x^2 + x - 3, \quad g(x) = x^3 - 2x^2 + x - 2, \quad h(x) = x^3 - 3x^2 + 3x - 9$.

3 Complex numbers

3.1 Why do we need complex numbers?

Example 3.1. We need complex numbers because they allow us to find the roots of every polynomial equation with real coefficients, even equations like $x^2 + 1 = 0$ (see exercise 3.6). These roots are then used to study matrices of linear operators during the course.

Example 3.2. Sometimes complex numbers help us find real roots of polynomial equations. This is particularly true for third-degree equations, as shown in Cardano's formula. If we closely analyze the formula, we can observe that even in cases where the roots are real, we may need to extract square roots of negative numbers at some point during the calculation.

Example 3.3. Complex numbers are used in signal processing, where they help study periodic functions like $\sin(\omega t)$ and $\cos(\omega t)$, since $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$.

Example 3.4. Complex numbers are widely used in quantum computing, and without them, it would be impossible to construct and use a quantum computer.

3.2 Field of complex numbers

For some reason we will describe later (see theorem 3.11), we will call the set of complex numbers a field.

Definition 3.1. The field of <u>*complex numbers*</u>, denoted by $\mathbb{C}$, is the set of all ordered pairs of real numbers

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(a, b) \mid a, b \in \mathbb{R}\}$$

with addition and multiplication defined as follows:

$$(a, b) + (c, d) = (a + c, b + d) \text{ for all } (a, b),\ (c, d) \in \mathbb{R}^2,$$

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc) \text{ for all } (a, b),\ (c, d) \in \mathbb{R}^2.$$

Theorem 3.2. $\mathbb{C}$ from definition 3.1 satisfies the following properties:

1) For any $(a, b), (c, d) \in \mathbb{R}^2$, we have $(a, b) + (c, d) = (c, d) + (a, b)$ (addition is commutative);

2) For any $(a, b), (c, d), (e, f) \in \mathbb{R}^2$, we have $((a, b) + (c, d)) + (e, f) = (a, b) + ((c, d) + (e, f))$ (addition is associative);

3) There exists $0 = (0, 0) \in \mathbb{C}$ such that for any $(a, b) \in \mathbb{R}^2$, we have $(0, 0) + (a, b) = (a, b)$ (existence of additive identity);

4) For any $(a, b) \in \mathbb{R}^2$ there exists $-(a, b) = (-a, -b)$, such that $(a, b) + (-(a, b)) = (0, 0)$ (existence of additive inverse);

5) For any $(a, b), (c, d), (e, f) \in \mathbb{R}^2$, we have $(a, b) \cdot ((c, d) + (e, f)) = (a, b) \cdot (c, d) + (a, b) \cdot (e, f)$ (multiplication is distributive over addition);

6) For any $(a, b), (c, d) \in \mathbb{R}^2$, we have $(a, b) \cdot (c, d) = (c, d) \cdot (a, b)$ (multiplication is commutative);

7) For any $(a, b), (c, d), (e, f) \in \mathbb{R}^2$, we have $(a, b) \cdot ((c, d) \cdot (e, f)) = ((a, b) \cdot (c, d)) \cdot (e, f)$ (multiplication is associative);

8) There exists $1 = (1, 0) \in \mathbb{C}$ such that for any $(a, b) \in \mathbb{R}^2$, we have $(1, 0) \cdot (a, b) = (a, b)$ (existence of multiplicative identity);

9) For any $(a, b) \in \mathbb{R}^2 \backslash \{(0, 0)\}$ there exists $(a, b)^{-1} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$, such that $(a, b) \cdot (a, b)^{-1} = (1, 0)$ (existence of multiplicative inverse).

Proof. Check all conditions from the definition:

1) For any $(a, b), (c, d) \in \mathbb{R}^2$, we have:

$$\begin{aligned}(a, b) + (c, d) &= (a + c, b + d)\\ &= (c + a, d + b)\\ &= (c, d) + (a, b).\end{aligned}$$

2) For any $(a,b),(c,d),(e,f) \in \mathbb{R}^2$, we have:

$$\begin{aligned}((a,b)+(c,d))+(e,f) &= (a+c,b+d)+(e,f)\\ &= ((a+c)+e,(b+d)+f)\\ &= (a+(c+e),b+(d+f))\\ &= (a,b)+(c+e,d+f)\\ &= (a,b)+((c,d)+(e,f)).\end{aligned}$$

3) Let $\mathbb{C} \ni 0 = (0,0) \in \mathbb{R}^2$, then for any $(a,b) \in \mathbb{R}^2$, we have:

$$\begin{aligned}(0,0)+(a,b) &= (0+a,0+b)\\ &= (a,b).\end{aligned}$$

4) For any $(a,b) \in \mathbb{R}^2$ let $-(a,b) = (-a,-b)$, then

$$\begin{aligned}(a,b)+(-(a,b)) &= (a,b)+(-a,-b)\\ &= (a+(-a),b+(-b))\\ &= (0,0).\end{aligned}$$

5) For any $(a,b),(c,d),(e,f) \in \mathbb{R}^2$, we have:

$$\begin{aligned}(a,b)\cdot((c,d)+(e,f)) &= (a,b)\cdot(c+e,d+f)\\ &= (a(c+e)-b(d+f),a(d+f)+b(c+e))\\ &= (ac+ae-bd-bf,ad+af+bc+be)\\ &= (ac-bd+ae-bf,ad+bc+af+be)\\ &= (ac-bd,ad+bc)+(ae-bf,af+be)\\ &= (a,b)\cdot(c,d)+(a,b)\cdot(e,f).\end{aligned}$$

6) For any $(a,b),(c,d) \in \mathbb{R}^2$, we have:

$$\begin{aligned}(a,b)\cdot(c,d) &= (ac-bd,ad+bc)\\ &= (ca-db,cb+da)\\ &= (c,d)\cdot(a,b).\end{aligned}$$

7) For any $(a,b),(c,d),(e,f) \in \mathbb{R}^2$, we have:

$$\begin{aligned}(a,b)\cdot((c,d)\cdot(e,f)) &= (a,b)\cdot(ce-df,cf+de)\\ &= (a(ce-df)-b(cf+de),a(cf+de)+b(ce-df))\\ &= (ace-adf-bcf-bde,acf+ade+bce-bdf)\\ &= (ace-bde-adf-bcf,acf-bdf+ade+bce)\\ &= ((ac-bd)e-(ad+bc)f,(ac-bd)f+(ad+bc)e)\\ &= (ac-bd,ad+bc)\cdot(e,f)\\ &= ((a,b)\cdot(c,d))\cdot(e,f).\end{aligned}$$

8) Let $\mathbb{C} \ni 1 = (1,0) \in \mathbb{R}^2$, then for any $(a,b) \in \mathbb{R}^2$, we have:

$$\begin{aligned}(1,0)\cdot(a,b) &= (1a-0b,1b+0a)\\ &= (a,b).\end{aligned}$$

9) For any $(a,b) \in \mathbb{R}^2\backslash\{(0,0)\}$ let $(a,b)^{-1} = \left(\frac{a}{a^2+b^2},\frac{-b}{a^2+b^2}\right)$, then

$$\begin{aligned}(a,b)\cdot(a,b)^{-1} &= (a,b)\cdot\left(\frac{a}{a^2+b^2},\frac{-b}{a^2+b^2}\right)\\ &= \left(\frac{a^2}{a^2+b^2}-\frac{b(-b)}{a^2+b^2},\frac{a(-b)}{a^2+b^2}+\frac{ba}{a^2+b^2}\right)\\ &= (1,0).\end{aligned}$$

$\square$

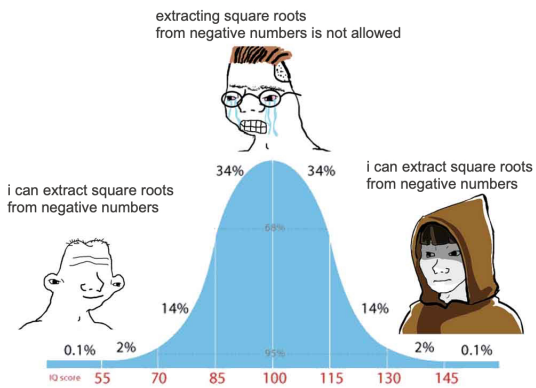
We can identify the subset $S = \{(a,0) \mid a \in \mathbb{R}\} \subset \mathbb{R}^2$ with real numbers. Indeed, if $a \in \mathbb{R}$ corresponds to $(a,0)$, then addition and multiplication of these complex numbers work as in the reals:

$$(a,0) + (b,0) = (a+b,0)$$
$$(a,0) \cdot (b,0) = \big(ab - 0^2, a0 + 0b\big) = (ab,0).$$

Definition 3.3. The element $(0,1) \in \mathbb{C}$ is called <u>*the imaginary unit*</u> and denoted $i = (0,1)$.

Exercise 3.5. Prove that $i^2 = -1 = (-1,0)$.

Exercise 3.6. Prove that the equation $x^2 + 1 = 0$ has exactly two solutions $\pm i$ in $\mathbb{C}$.



Definition 3.4. The representation $z = (a,b)$ is called the <u>*Cartesian form*</u> of a complex number z (named after René Descartes, *lat.* Renatus Cartesius).

Lemma 3.5. Any complex number $z = (a,b)$ can be uniquely represented in the form $z = a + bi$.

Proof. Since $i = (0,1)$, we have $z = (a,b) = (a,0) + (b,0) \cdot (0,1) = a + bi$, so the representation exists.

If $a + bi = a' + b'i$ then

$$(a,b) = (a,0) + (b,0) \cdot (0,1) = a + bi = a' + b'i = (a',0) + (b',0) \cdot (0,1) = (a',b').$$

By the definition of ordered pair, from the equality $(a,b) = (a',b')$ we have $a = a'$ and $b = b'$, so the representation is unique. $\square$

Definition 3.6. The form $a + bi$ is called the <u>*algebraic form*</u> of a complex number z.

Exercise 3.7. Check that all properties from theorem 3.2 hold true in algebraic form.

Definition 3.7. Let $z = a + bi$ be a complex number, then the <u>*real part*</u> of z is

$$\mathrm{Re}(z) = \mathrm{Re}(a + bi) = a,$$

and the imaginary part of z is

$$\mathrm{Im}(z) = \mathrm{Im}(a + bi) = b.$$

Remark 3.8. Both $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ are real and $\mathrm{Re}(z) + i\,\mathrm{Im}(z) = z$ for any complex number z.

Definition 3.8. The <u>*absolute value*</u> (or <u>*modulus*</u>) of a complex number $z = a + bi$ is

$$|z| = \sqrt{a^2 + b^2}.$$

Remark 3.9. For real numbers, the absolute value in the real and complex senses is the same.

Exercise 3.10. Find $|z|$ where $z = 1, i, -i, 1 + i$.

3.3 Geometric interpretation

In the geometric interpretation, a complex number $z = a + bi$ is represented as a vector in a Cartesian coordinate system, extending from the point $(0, 0)$ to the point $(a, b) = (\mathrm{Re}(z), \mathrm{Im}(z))$.

The addition of complex numbers corresponds to the addition of vectors.

Exercise 3.11. Prove that complex numbers form a vector space over the real numbers. Find a basis and the dimension of this vector space.

3.4 * Fields

Definition 3.9. A *binary operation* on a set S is a mapping of the elements of the Cartesian product $S \times S$ to S:

$$f\colon S \times S \to S.$$

Definition 3.10. A *field* is a triple $(F, +, \cdot)$, where F is a set, and

$$+\colon F \times F \to F,$$

$$\cdot\colon F \times F \to F$$

are binary operations (written $(a, b) \mapsto a + b$ and $(a, b) \mapsto a \cdot b$ and called addition and multiplication, respectively) such that the following conditions are satisfied for any $a, b, c \in F$:

1. $a + b = b + a$ (commutativity of addition),
2. $(a + b) + c = a + (b + c)$ (associativity of addition),
3. $0 \in F$ such that $0 + a = a$ (existence of additive identity),
4. for any $a \in F$ there is $-a \in F$ such that $a + (-a) = 0$ (existence of additive inverse),
5. $a \cdot b = b \cdot a$ (commutativity of multiplication),
6. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ (associativity of multiplication),
7. there is $1 \in F \backslash \{0\}$ such that $1 \cdot a = a$ (existence of multiplicative identity),
8. if $a \in F \backslash \{0\}$ then there is $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$ (existence of multiplicative inverse),
9. $a \cdot (b + c) = a \cdot b + a \cdot c$ (distributivity of multiplication over addition).

Remark 3.12. We normally omit the multiplication sign $\cdot$, especially when it is clear from the context.

Example 3.13. a) $\mathbb{R}$ — field of real numbers, operations: regular addition and multiplication of numbers,

b) $\mathbb{Q}$ — field of rationals, operations are the same,

c) $\mathbb{F}_2 = (\{0, 1\}, +_{\bmod 2}, \cdot_{\bmod 2})$.

Exercise 3.14. Check that fields from the previous example a), b), c) are actually fields. Build addition and multiplication tables for $\mathbb{F}_2$.

Theorem 3.11. $\mathbb{C}$ from definition 3.1 is actually a field.

Proof. Follows immediately from theorem 3.2. □

Exercise 3.15. Let $\mathbb{F}$ be a field. Prove that

a) the additive identity $0 \in \mathbb{F}$ is unique;

b) the multiplicative identity of $1 \in \mathbb{F}$ is unique;

c) $\forall x \in \mathbb{F}\ \exists!\ -x \in \mathbb{F}$ (the additive inverse is unique for every field element);

d) $\forall x \in \mathbb{F} \backslash \{0\}\ \exists!\ x^{-1} \in \mathbb{F}$;

e) $\forall x \in \mathbb{F}\ x \cdot 0 = 0 \cdot x = 0$;

f) $\forall x \in \mathbb{F}\ (-1) \cdot x = x \cdot (-1) = -x$.

Definition 3.12. If $\mathbb{F} = (F, +, \cdot)$ is a field and a subset S of F satisfies conditions 1)–9) from the definition of a field with the same operations as in $\mathbb{F}$, then $\mathbb{S} = (S, +, \cdot)$ is called a subfield of $\mathbb{F}$.

Example 3.16. $\mathbb{Q}$ is a subfield of $\mathbb{R}$.

Lemma 3.13. Let $S = \{(a, 0) \mid a \in \mathbb{R}\} \subset \mathbb{R}^2$. Then S, with the operations inherited from $\mathbb{C}$, $(S, +, \cdot)$, is a subfield of $\mathbb{C}$, isomorphic to $\mathbb{R}$.

3.5 Complex conjugate

Definition 3.14. The <u>*complex conjugate*</u> of a complex number $z = a + bi$, denoted $\bar{z}$, is defined to be

$$\bar{z} = a - bi.$$

The geometric interpretation of complex conjugation is reflection with respect to the x-axis.

Theorem 3.15. For any $z, z_1, z_2 \in \mathbb{C}$, the complex conjugate satisfies the following properties:

1. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$;
2. $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$;
3. $(\bar{z} = z) \Leftrightarrow (\mathrm{Im}(z) = 0)$;
4. $|\bar{z}| = |z|$;
5. $\bar{\bar{z}} = z$;
6. $\bar{z} \cdot z = |z|^2$.

Exercise 3.17. Prove theorem 3.15.

Exercise 3.18. Prove that:

a) a complex number z is real if and only if $\overline{z} = z$;

b) a nonzero complex number z is purely imaginary if and only if $\overline{z} = -z$.

Exercise 3.19. Prove that if z is a root of a polynomial with real coefficients, then its complex conjugate $\overline{z}$ is also a root of the polynomial.

Exercise 3.20. Prove that every real polynomial can be represented as a product of linear polynomials with real coefficients and quadratic polynomials with real coefficients and negative discriminant.

3.6 Polar coordinates

Each point in the plane (a, b), except for the origin $(0, 0)$, is uniquely determined by its polar coordinates (r, φ), where r represents the distance from the point to the origin, and φ is the angle between the positive x-axis and the radius vector of the point (a, b), measured counterclockwise. The angle is determined up to an integer multiple of 2π, i.e., $2\pi k$ with $k \in \mathbb{Z}$. This angle is referred to as the *argument* of the point.

Note that for the point $(0, 0)$ the argument is undefined.

Exercise 3.21. Find the coordinate transformation formulas from Cartesian to polar coordinates and vice versa.

Since there is a correspondence between complex numbers and points in $\mathbb{R}^2$, where a complex number $a + bi$ corresponds to the point (a, b), we can utilize polar coordinates to analyze problems involving complex numbers. By doing so, we can use geometric concepts to resolve algebraic problems regarding complex numbers, and vice versa. In this context, we define the argument of a complex number $a + bi$ as the argument of the corresponding point (a, b) and denote it $\arg(a + bi)$.

Note that with this definition, the argument of a complex number is not uniquely determined.

3.7 Absolute value properties

Lemma 3.16. 1. $\forall z, w \in \mathbb{C}\ |zw| = |z||w|$,

2. $\forall z \in \mathbb{C}\backslash\{0\}\ |z^{-1}| = |z|^{-1}$,

3. $\forall z \in \mathbb{C}, w \in \mathbb{C}\backslash\{0\}\ \left|\frac{z}{w}\right| = \frac{|z|}{|w|}$.

Proof. 1. By lemma 3.15, we have

$$|zw| = \sqrt{(zw)\overline{(zw)}} = \sqrt{zw\overline{zw}} = \sqrt{z\bar{z}}\sqrt{w\bar{w}} = |z||w|.$$

2. $1 = |1| = |zz^{-1}| = |z||z^{-1}|$, then $|z^{-1}| = \frac{1}{|z|}$.

3. $\left|\frac{z}{w}\right| = |zw^{-1}| = |z||w^{-1}| = |z||w|^{-1} = \frac{|z|}{|w|}$.

□

Lemma 3.17. $\forall z, w \in \mathbb{C}\ |z + w| \leqslant |z| + |w|$.

Proof. Using the geometric interpretation of complex numbers and their sums, consider a triangle with sides $|z|, |w|, |z + w|$. Applying the triangle inequality to this triangle, we prove the lemma. □

Exercise 3.22. Prove that for $z, w \in \mathbb{C}\ |w| - |z| \leqslant |w \pm z| \leqslant |w| + |z|$.

3.8 Trigonometric form

Using the polar coordinates r, φ of the point (a, b) representing the complex number $a + bi$, we have

$$a + bi = r\cos\varphi + r\sin\varphi i = r(\cos\varphi + i\sin\varphi),$$

which is called <u>*the trigonometric form*</u> of a complex number.

Exercise 3.23. Let $z = r(\cos\varphi + i\sin\varphi),\ r > 0$. Find the trigonometric form for

a) $-z$,

b) $\bar{z}$.

Theorem 3.18. Let $z_1 = r_1(\cos\varphi_1 + i\sin\varphi_1)$ and $z_2 = r_2(\cos\varphi_2 + i\sin\varphi_2)$ be two nonzero complex numbers (where $r_1, r_2 > 0$). Then

1. $z_1 \cdot z_2 = r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2))$;

2. $z_1 \cdot z_2^{-1} = \frac{r_1}{r_2}(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2))$.

Proof. 1.

$$\begin{gathered} z_1 \cdot z_2 = (r_1(\cos\varphi_1 + i\sin\varphi_1))(r_2(\cos\varphi_2 + i\sin\varphi_2)) = \\ = r_1 r_2((\cos\varphi_1\cos\varphi_2 - \sin\varphi_1\sin\varphi_2) + i(\cos\varphi_1\sin\varphi_2 + \sin\varphi_1\cos\varphi_2)) = \\ = r_1 r_2(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)). \end{gathered}$$

2.

$$\begin{gathered} z_1 \cdot z_2^{-1} = \frac{z_1 \cdot \overline{z_2}}{z_2 \cdot \overline{z_2}} = \frac{z_1 \cdot \overline{z_2}}{|z_2|^2} = \\ = \frac{1}{r_2^2}(r_1(\cos\varphi_1 + i\sin\varphi_1))\left(\overline{r_2(\cos\varphi_2 + i\sin\varphi_2)}\right) = \\ = \frac{1}{r_2^2}(r_1(\cos\varphi_1 + i\sin\varphi_1))(r_2(\cos\varphi_2 - i\sin\varphi_2)) = \\ = \frac{r_1}{r_2}((\cos\varphi_1\cos\varphi_2 + \sin\varphi_1\sin\varphi_2) + i(-\cos\varphi_1\sin\varphi_2 + \sin\varphi_1\cos\varphi_2)) = \\ = \frac{r_1}{r_2}(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)). \end{gathered}$$

□

Theorem 3.19. (Uniqueness of the trigonometric form.) If $0 \neq z = a + bi \in \mathbb{C}$ and

$$z = r_1 (\cos\varphi_1 + i \sin\varphi_1) = r_2 (\cos\varphi_2 + i \sin\varphi_2),$$

where $\mathbb{R} \ni r_1 > 0, \mathbb{R} \ni r_2 > 0$, then

$$r_1 = r_2 \quad \text{and} \quad \varphi_1 - \varphi_2 = 2\pi k, \quad k \in \mathbb{Z}.$$

Proof. From the uniqueness of the algebraic form, we have

$$a = r_1 \cos\varphi_1 = r_2 \cos\varphi_2 \quad \text{and} \quad b = r_1 \sin\varphi_1 = r_2 \sin\varphi_2.$$

Squaring both equations and adding them together, we obtain

$$r_1^2 = r_1^2 \left(\cos^2\varphi_1 + \sin^2\varphi_1\right) = r_2^2 \left(\cos^2\varphi_2 + \sin^2\varphi_2\right) = r_2^2.$$

Since $r_1 > 0$ and $r_2 > 0$, it follows that $r_1 = r_2$. Therefore, $\cos\varphi_1 = \cos\varphi_2$ and $\sin\varphi_1 = \sin\varphi_2$, which implies $\varphi_1 - \varphi_2 = 2\pi k,\ k \in \mathbb{Z}$.

□

3.9 De Moivre's formula. Roots

Theorem 3.20. (De Moivre's formula.) If $z = r(\cos\varphi + i\sin\varphi)$ is a nonzero complex number ($r > 0$), then for any $n \in \mathbb{Z}$ we have:

$$z^n = r^n(\cos(n\varphi) + i\sin(n\varphi)).$$

Exercise 3.24. Prove theorem 3.20. (Hint: you may prove this by induction.)

Definition 3.21. A complex number u is said to be an <u>n-th root</u> of a complex number z if $u^n = z$.

Remark 3.25. In terms of this definition, there are two square roots of the number 4: 2 and -2. The principal square root of a positive real number is unique; for instance, $\sqrt{4} = 2$.

We denote by $\sqrt[n]{z}$ the set of all n-th roots of a complex number z:

$$\sqrt[n]{z} = \{u \in \mathbb{C} \mid u^n = z\}.$$

For example:

$$\sqrt[3]{1} = \sqrt[3]{1 + 0i} = \left\{1 + 0i, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i\right\}.$$

Theorem 3.22. Let $z = r(\cos\varphi + i\sin\varphi)$ be a complex number. If $z = 0$ then $\sqrt[n]{z} = \{0\}$. If $z \neq 0$ then

$$\sqrt[n]{z} = \left\{\sqrt[n]{r}\left(\cos\left(\frac{\varphi + 2\pi k}{n}\right) + i\sin\left(\frac{\varphi + 2\pi k}{n}\right)\right) \mid k \in \{0, 1, \ldots, n-1\}\right\},$$

where $\sqrt[n]{r}$ is the principal root (positive) from a positive real number.

Proof. We will look for the solutions w of the equation $w^n = z$ in trigonometric form:

$$w = \rho(\cos\theta + i\sin\theta), \quad \rho > 0.$$

Then, by De Moivre's formula,

$$w^n = \rho^n(\cos n\theta + i\sin n\theta) = r(\cos\varphi + i\sin\varphi) = z,$$

which implies $\rho^n = r$. By the uniqueness of the trigonometric form, $\rho = \sqrt[n]{r}$, and $n\theta = \varphi + 2\pi k, k \in \mathbb{Z}$. There will be exactly n distinct roots when $k = 0, 1, 2, \ldots, n-1$:

$$w_k = \sqrt[n]{r}\left(\cos\frac{\varphi + 2\pi k}{n} + i\sin\frac{\varphi + 2\pi k}{n}\right): \quad k = 0, 1, 2, \ldots, n-1.$$

□

Exercise 3.26. How many different n-th roots of a nonzero complex number exist?

Exercise 3.27. Describe the locus of all n-th roots of a nonzero complex number. Draw a sketch.

Joke. Is it true that $\sqrt[ns]{z^s} = \sqrt[n]{z}$ $(s > 1)$?

3.10 Fundamental theorem of algebra

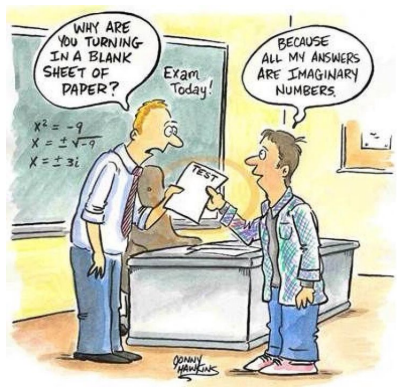
Theorem 3.23. Every polynomial of positive degree with complex coefficients has at least one complex root.

(* *Proof* can be found in Axler, Chapter 4, Factorization of Polynomials over $\mathbb{C}$.)

Corollary 3.24. Every polynomial of positive degree n with complex coefficients has exactly n complex roots, taking into account their multiplicity.

Exercise 3.28. Prove corollary 3.24.

Corollary 3.25. Every polynomial of positive degree n with real coefficients has exactly n complex roots, taking into account their multiplicity.



3.11 Problems

Problem 3.1. Find the algebraic form of an expression:

a) $(2+i)(3-i)+(2+3i)(3+4i)$;

b) $(2+i)(3+7i)-(1+2i)(5+3i)$;

c) $(4+i)(5+3i)-(3+i)(3-i)$;

d) $\frac{(5+i)(7-6i)}{3+i}$

e) $\frac{(5+i)(3+5i)}{2i}$;

f) $\frac{(1+3i)(8-i)}{(2+i)^2}$;

g) $\frac{(2+i)(4+i)}{1+i}$

h) $\frac{(3-i)(1-4i)}{2-i}$;

i) $(2+i)^3+(2-i)^3$;

j) $(3+i)^3-(3-i)^3$;

k) $\frac{(1+i)^5}{(1-i)^3}$

l) $\left(-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i\right)^3$.

Problem 3.2. Calculate

a) i^{77},

b) i^{98},

c) i^{-57},

d) i^n,

where n is an integer (do not forget to consider negative n).

Problem 3.3. Prove the equalities:

a) $(1+i)^{8n}=2^{4n}, \quad n\in\mathbb{Z}$;

b) $(1+i)^{4n}=(-1)2^{2n}, \quad n\in\mathbb{Z}$.

Problem 3.4. Solve the system of equations:

a) $\begin{cases} (1+i)z_1 + (1-i)z_2 = 1+i, \\ (1-i)z_1 + (1+i)z_2 = 1+3i; \end{cases}$

b) $\begin{cases} iz_1 + (1+i)z_2 = 2+2i, \\ 2iz_1 + (3+2i)z_2 = 5+3i; \end{cases}$

c) $\begin{cases} (1-i)z_1 - 3iz_2 = -i, \\ 2z_1 - (3+3i)z_2 = 3-i; \end{cases}$

d) $\begin{cases} 2z_1 - (2+i)z_2 = -i, \\ (4-2i)z_1 - 5z_2 = -1-2i; \end{cases}$

e) $\begin{cases} x + iy - 2z = 10, \\ x - y + 2iz = 20, \\ ix + 3iy - (1+i)z = 30. \end{cases}$

Problem 3.5. Find all real x and y such that:

a) $(2+i)x + (1+2i)y = 1-4i$;

b) $(3+2i)x + (1+3i)y = 4-9i$.

Problem 3.6. Solve the equation:

a) $z^2 = i$;

b) $z^2 = 3-4i$;

c) $z^2 = 5-12i$;

d) $z^2 - (1+i)z + 6 + 3i = 0$;

e) $z^2 - 5z + 4 + 10i = 0$;

f) $z^2 + (2i-7)z + 13 - i = 0$;

g) $x^5 - 4x^4 + 4x^3 - x^2 + 4x - 4 = 0$;

h) $x^5 + x^4 + 2x + 2 = 0$;

Problem 3.7. Prove that:

a) the product of two complex numbers is real if and only if one of them differs from the complex conjugate of the other by a real factor;

b) the sum and product of two complex numbers are real if and only if the given numbers are either conjugates or both real.

Problem 3.8. Find all complex numbers that are conjugate to a) their square, b) their cube.

Problem 3.9. Prove that if the complex numbers $z_1, z_2, \ldots, z_n$ can be obtained from a finite number of operations of addition, subtraction, multiplication, and division, and the result is the number z, then the complex numbers $\overline{z_1}, \overline{z_2}, \ldots, \overline{z_n}$ obtained by applying the same operations are equal to $\overline{z}$.

Problem 3.10. Find the trigonometric form for the following numbers:

a) 5;

b) i;

c) -2;

d) $-3i$;

e) $1+i$;

f) $1-i$;

g) $1+i\sqrt{3}$;

h) $-1+i\sqrt{3}$;

i) $-1-i\sqrt{3}$;

j) $1-i\sqrt{3}$;

k) $\sqrt{3}+i$;

l) $-\sqrt{3}+i$;

m) $-\sqrt{3}-i$;

n) $\sqrt{3}-i$;

o) $1+i\frac{\sqrt{3}}{3}$;

p) $2+\sqrt{3}+i$;

q) $1-(2+\sqrt{3})i$;

r) $\cos\alpha - i\sin\alpha$;

s) $\sin\alpha + i\cos\alpha$;

t) $\frac{1+i\operatorname{tg}\alpha}{1-i\operatorname{tg}\alpha}$;

u) $1+\cos\varphi + i\sin\varphi, \quad \varphi \in [-\pi, \pi]$;

v) $\dfrac{\cos\varphi + i\sin\varphi}{\cos\psi + i\sin\psi}$.

Problem 3.11. Calculate each expression and find its algebraic form:

a) $(1+i)^{1000}$;

b) $(1+i\sqrt{3})^{150}$;

c) $(\sqrt{3}+i)^{30}$;

d) $\left(1+\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^{24}$;

e) $(2-\sqrt{3}+i)^{12}$;

f) $\left(\frac{1-i\sqrt{3}}{1+i}\right)^{12}$;

g) $\left(\frac{\sqrt{3}+i}{1-i}\right)^{30}$;

h) $\frac{(-1+i\sqrt{3})^{15}}{(1-i)^{20}} + \frac{(-1-i\sqrt{3})^{15}}{(1+i)^{20}}$.

Problem 3.12. For $n \in \mathbb{Z}$ find the algebraic form for:

a) $(1+i)^n$;

b) $\left(\frac{1-i\sqrt{3}}{2}\right)^n$;

c) $\left(\frac{1-i\operatorname{tg}\alpha}{1+i\operatorname{tg}\alpha}\right)^n$;

d) $(1+\cos\varphi + i\sin\varphi)^n$.

Problem 3.13. Express the following functions as polynomials in terms of $\sin x$ and $\cos x$:

a) $\sin 4x$;

b) $\cos 4x$;

c) $\sin 5x$;

d) $\cos 5x$.

Problem 3.14. Calculate:

a) $\sqrt[6]{i}$;

b) $\sqrt[10]{512(1-i\sqrt{3})}$;

c) $\sqrt[8]{2\sqrt{2}(1-i)}$;

d) $\sqrt[3]{1}$;

e) $\sqrt[4]{1}$;

f) $\sqrt[6]{1}$;

g) $\sqrt[3]{i}$;

h) $\sqrt[4]{-4}$;

i) $\sqrt[6]{64}$;

j) $\sqrt[8]{16}$;

k) $\sqrt[6]{-27}$;

l) $\sqrt[4]{8\sqrt{3}i-8}$;

m) $\sqrt[4]{-72(1-i\sqrt{3})}$;

n) $\sqrt[3]{1+i}$;

o) $\sqrt[3]{2-2i}$;

p) $\sqrt[4]{-\frac{18}{1+i\sqrt{3}}}$;

q) $\sqrt[4]{\frac{7-2i}{1+i\sqrt{2}} + \frac{4+14i}{\sqrt{2}+2i} - (8-2i)}$;

r) $\sqrt[3]{\frac{1-5i}{1+i} - 5\frac{1+2i}{2-i} + 2}$;

s) $\sqrt[4]{\frac{-2+2\sqrt{3}i}{2+i\sqrt{5}} - 5\frac{\sqrt{3}+i}{2\sqrt{5}+5i}}$.

3.12 Additional problems

Problem 3.15. Find the sum and product of all complex nth roots of 1.

Problem 3.16. Find the fifth roots of unity in two ways and express the following numbers in radicals:

a) $\cos\frac{2\pi}{5}$;

b) $\sin\frac{2\pi}{5}$;

c) $\cos\frac{4\pi}{5}$;

d) $\sin\frac{4\pi}{5}$.

Problem 3.17. Find sums:

a) $1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \ldots$;

b) $\binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \binom{n}{7} + \ldots$;

c) $1 + \binom{n}{4} + \binom{n}{8} + \ldots$;

d) $\binom{n}{1} + \binom{n}{5} + \binom{n}{9} + \ldots$.

Hint: why is this a problem on complex numbers? :)

Problem 3.18. Prove the following equalities:

a) $\cos x + \cos 2x + \ldots + \cos nx = \frac{\sin\frac{nx}{2}\cos\frac{(n+1)x}{2}}{\sin\frac{x}{2}} \quad (x \neq 2k\pi, k \in \mathbb{Z})$;

b) $\sin x + \sin 2x + \ldots + \sin nx = \frac{\sin\frac{nx}{2}\sin\frac{(n+1)x}{2}}{\sin\frac{x}{2}} \quad (x \neq 2k\pi, k \in \mathbb{Z})$;

c) $\cos\frac{\pi}{n} + \cos\frac{3\pi}{n} + \cos\frac{5\pi}{n} + \ldots + \cos\frac{(2n-1)\pi}{n} = 0$;

d) $\sin\frac{\pi}{n} + \sin\frac{3\pi}{n} + \sin\frac{5\pi}{n} + \ldots + \sin\frac{(2n-1)\pi}{n} = 0$;

e) $\frac{1}{n}\sum_{k=0}^{n-1}(x+\varepsilon_k y)^n = x^n + y^n \quad (\varepsilon_0, \varepsilon_1, \ldots, \varepsilon_{n-1}$ are nth roots of unity).

4 Linear operators

4.1 Direct sum of subspaces

From now on, we assume that the field $\mathbb{F}$ is either the field of real numbers or the field of complex numbers.

We recall that vectors $v_1, \dots, v_n$ in a vector space are linearly independent if from the relation $\alpha_1 v_1 + \dots + \alpha_n v_n = 0$ it follows that $\alpha_1 = \dots = \alpha_n = 0$; equivalently, every vector in the span of $\{v_1, \dots, v_n\}$ is represented uniquely as a linear combination of $v_1, \dots, v_n$. Now we will try to generalize this definition to subspaces.

Let V be a vector space over a field $\mathbb{F}$, and let $U_1, \dots, U_k$ be subspaces of V. Consider their sum

$$U = U_1 + \dots + U_k = \{u_1 + \dots + u_k \mid u_i \in U_i\}.$$

Clearly, this is a subspace of V. In general, a vector $u \in U$ may have more than one representation as a sum of vectors from $U_1, \dots, U_k$.

Example 4.1. Consider the vector space $\mathbb{R}^3$ and the subspaces $U_1 = \{(x, y, 0)\}$ and $U_2 = \{(0, a, b)\}$. Then the sum $U_1 + U_2$ is the whole space $\mathbb{R}^3$. Vector $(1, 3, 1)$ can be represented as $(1, 1, 0) + (0, 2, 1) = (1, 2, 0) + (0, 1, 1)$ in two different ways.

Definition 4.1. The sum of subspaces $U_1, \dots, U_k$ of a vector space V is called <u>*direct*</u> if any vector $u \in U$ can be represented as a sum of vectors from $U_1, \dots, U_k$ in a unique way. In this case we write $U = U_1 \oplus \dots \oplus U_k$.

Lemma 4.2. The sum of subspaces $U_1, \dots, U_k$ of a vector space V is direct if and only if the zero vector is represented as a sum $0 = u_1 + \dots + u_k$ in a unique way, i.e. if and only if $u_1 = \dots = u_k = 0$.

Proof. The "if" part is obvious. For the "only if" part, assume that there is a non-zero vector $w \in U$ such that $w = u_1 + \dots + u_k = u'_1 + \dots + u'_k$ in two different ways. Then $(u_1 - u'_1) + \dots + (u_k - u'_k) = 0$, where $u_i - u'_i \in U_i$ and at least one of the vectors $u_i - u'_i \neq 0$, which contradicts the assumption that 0 has a unique representation. □

Exercise 4.2. Prove that for two subspaces U_1 and U_2 of a vector space V, the sum $U = U_1 \oplus U_2$ is direct if and only if $U_1 \cap U_2 = \{0\}$.

Exercise 4.3. Provide an example of a vector space and three of its subspaces U_1, U_2, U_3 such that $U_1 \cap U_2 = \{0\}$, $U_2 \cap U_3 = \{0\}$, $U_3 \cap U_1 = \{0\}$, but the sum $U = U_1 + U_2 + U_3$ is not direct.

Lemma 4.3. Prove that the sum of subspaces $U_1, \dots, U_k$ of a vector space V is direct if and only if

$$U_i \cap (U_1 + \dots + U_{i-1} + U_{i+1} + \dots + U_k) = \{0\} \quad \text{for all } i = 1, 2, \dots, k.$$

Proof. Assume first that the sum $U = U_1 \oplus \dots \oplus U_k$ is direct. Let $u \in U_i \cap (U_1 + \dots + U_{i-1} + U_{i+1} + \dots + U_k)$. Then u has two representations as a sum of vectors from U_j:

$$u = u_i \quad (u_i \in U_i), \qquad u = u_1 + \dots + u_{i-1} + u_{i+1} + \dots + u_k \quad (u_j \in U_j).$$

By uniqueness of representation in a direct sum, all components must be zero, whence $u = 0$. Thus,

$$U_i \cap (U_1 + \dots + U_{i-1} + U_{i+1} + \dots + U_k) = \{0\}$$

for every i.

Conversely, assume that for every i we have

$$U_i \cap (U_1 + \dots + U_{i-1} + U_{i+1} + \dots + U_k) = \{0\}.$$

Suppose $u_1 + \dots + u_k = 0$ with $u_i \in U_i$. Then

$$u_1 = -(u_2 + \dots + u_k) \in U_1 \cap (U_2 + \dots + U_k) = \{0\},$$

so $u_1 = 0$. Repeating the same argument for $i = 2, \dots, k$ gives $u_2 = \dots = u_k = 0$. Hence the representation of any vector in $U_1 + \dots + U_k$ is unique, and therefore the sum is direct. □

Lemma 4.4. Let $U = U_1 \oplus \dots \oplus U_k$ be a direct sum of subspaces $U_1, \dots, U_k$ of a vector space V. Let $e^i_1, \dots, e^i_{m_i}$ be a basis of U_i for each $i = 1, \dots, k$. Then the set of vectors $e^1_1, \dots, e^1_{m_1}, \dots, e^k_1, \dots, e^k_{m_k}$ is a basis of U.

Proof. It is clear that every vector $u \in U$ can be represented as a sum of vectors from the union of bases of U_i.

Prove linear independence of the set of vectors $e^1_1, \dots, e^1_{m_1}, \dots, e^k_1, \dots, e^k_{m_k}$. If $\alpha_1 e^1_1 + \dots + \alpha^1_{m_1} e^1_{m_1} + \dots + \alpha^k_1 e^k_1 + \dots + \alpha^k_{m_k} e^k_{m_k} = 0$, then $\alpha^i_1 e^i_1 + \dots + \alpha^i_{m_i} e^i_{m_i} = 0$ for all $i = 1, \dots, k$, because the sum is direct. Inside every U_i all the coefficients $\alpha^i_1, \dots, \alpha^i_{m_i}$ are zero, because the vectors $e^i_1, \dots, e^i_{m_i}$ are linearly independent. □

Corollary 4.5. $\dim(U_1 \oplus \dots \oplus U_k) = \dim(U_1) + \dots + \dim(U_k)$.

Exercise 4.4. Prove the converse: if $\dim(U_1 + \dots + U_k) = \dim(U_1) + \dots + \dim(U_k)$, then the sum is direct.

4.2 Linear operators: definition

The main purpose of a matrix is to be the matrix of a linear operator.

Definition 4.6. A linear operator $\mathcal{A}$ on a vector space V over a field F is a linear transformation that maps vectors in V to vectors in V. To be more precise, a linear operator is a map $\mathcal{A}: V \to V$ that satisfies the following properties for all vectors $v, w \in V$ and all scalars $\alpha \in \mathbb{F}$:

- $\mathcal{A}(v+w) = \mathcal{A}(v) + \mathcal{A}(w)$;
- $\mathcal{A}(\alpha v) = \alpha\,\mathcal{A}(v)$.

Remark 4.5. Later we will skip parentheses and write $\mathcal{A}\,v$ instead of $\mathcal{A}(v)$.

Example 4.6 (by a lazy mathematician). The zero operator $\mathcal{O}$ on any vector space V : $\mathcal{O}(v) = 0$ for all $v \in V$. The identity operator $I: V \to V$, $I(v) = v$ for all $v \in V$.

Exercise 4.7. Consider a map $\varphi: \mathbb{R}[x,3] \to \mathbb{R}[x,3]$, $\varphi(p(x)) = p''(x)(x^2+1) + p(5)$. Prove that φ is a linear operator.

Note that the set of all linear operators on a vector space V is a vector space itself (the zero vector is the zero operator); we denote it by $\mathcal{L}(V)$.

The matrix of a linear operator $\mathcal{A}: V \to V$ is a matrix that represents the linear operator with respect to a basis of V. Let $e = (e_1, e_2, \ldots, e_n)$ be a basis of a finite-dimensional vector space V. Then the matrix of $\mathcal{A}$ is the matrix A such that

$$\mathcal{A}\,e = eA.$$

If we expand this equation, we get

$$(\mathcal{A}\,e_1, \mathcal{A}\,e_2, \ldots, \mathcal{A}\,e_n) = \left(\sum_{i=1}^{n} A_{i1}e_i, \sum_{i=1}^{n} A_{i2}e_i, \ldots, \sum_{i=1}^{n} A_{in}e_i\right).$$

This means that the j-th column of the matrix A contains the coordinates of the image of the j-th basis vector under the linear operator. The size of this matrix is $n \times n$.

Remark 4.8. We will use calligraphic letters for linear operators, and regular letters for the corresponding matrices.

Exercise 4.9. Given the matrix A of a linear operator $\mathcal{A}$ on a vector space V with respect to the basis e of V, and a vector $v \in V$ with coordinates in the basis e, find the coordinates of the image of v under the linear operator $\mathcal{A}$ in the basis e.

Remark 4.10. We can define a linear operator on a vector space without specifying a basis (see examples above). But we cannot write down its matrix without fixing a basis.

Remark 4.11. It is easy to see that a linear operator (like any linear transformation) is uniquely determined by its values on the basis vectors (of some basis).

Exercise 4.12. Find the matrix of the linear operator φ from Exercise 4.7 with respect to the standard basis of $\mathbb{R}[x,3]$ (i.e. $1, x, x^2, x^3$).

Exercise 4.13. Find the matrices of $\mathcal{O}$ and I on a finite-dimensional vector space V. Do they depend on the basis?

Exercise 4.14. What does the matrix of a rotation of the plane by an angle $\varphi = \frac{\pi}{3}$ look like?

Exercise 4.15. Find any basis and the dimension of $\mathcal{L}(V)$, where V is a finite-dimensional vector space.

4.3 Properties of linear operators

In the previous section, we became acquainted with the matrix of a linear operator and even used it to solve problems. Fixing a basis and writing everything in matrix form can be useful for solving many problems. However, it is important to understand how the choice of basis affects the appearance of the matrix; that is, how the matrix of a linear operator changes when we switch to a different basis of the space V.

Lemma 4.7. Let $e = (e_1, e_2, \ldots, e_n)$ and $e' = (e'_1, e'_2, \ldots, e'_n)$ be two bases of a vector space V and let C be the change of basis matrix from e to e': $e' = eC$. Then the matrices of a linear operator $\mathcal{A}$ with respect to the bases e and e' are related by the formula

$$A' = C^{-1}AC.$$

Proof. We have $\mathcal{A}\,e' = e'A'$ and $\mathcal{A}\,e = eA$. Since $e' = eC$, we have $\mathcal{A}\,(eC) = eCA'$. By linearity of $\mathcal{A}$, we have $\mathcal{A}\,(eC) = (\mathcal{A}\,e)C$. Therefore, $\mathcal{A}\,e = eCA'C^{-1}$, which means that $A = CAC^{-1}$ and $A' = C^{-1}AC$. □

Since for every square matrix A and invertible matrix C of the same size we have $\det(A) = \det(CAC^{-1}) = \det(C)\det(A)\det(C)^{-1} = \det(A)$, the following definition is well-defined.

Definition 4.8. The <u>*determinant*</u> of a linear operator $\mathcal{A}: V \to V$ is the determinant of the matrix of $\mathcal{A}$ with respect to some (and hence any) basis of V.

Exercise 4.16. Find determinants of the zero operator and the identity operator on a finite-dimensional vector space V; rotation by an angle $\varphi = \frac{\pi}{3}$ on the plane.

Definition 4.9. Since the rank of a matrix does not change under multiplication by an invertible matrix, we can define the <u>*rank*</u> of a linear operator as the rank of the matrix of the operator with respect to some (and hence any) basis of V.

Definition 4.10. The <u>*trace*</u> of a square matrix A of size $n \times n$ is the sum of the elements on the main diagonal:

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Exercise 4.17. Is it true that $\operatorname{tr}(AB) = \operatorname{tr}(A)\operatorname{tr}(B)$? Prove it or provide a counterexample.

Surprisingly, the trace is invariant under change of basis as well! But to prove it we need to prove the following auxiliary lemma first.

Lemma 4.11. Let A and B be two square matrices of size $n \times n$. Then $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. We have $\operatorname{tr}(AB) = \sum_{i=1}^{n}(AB)_{ii} = \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}B_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{n} B_{ji}A_{ij} = \operatorname{tr}(BA)$. □

Exercise 4.18. Do there exist two square matrices A and B such that $AB - BA = I$? (Hint: why do we encounter this exercise at this point?)

Corollary 4.12. The trace is invariant under change of basis.

Proof. We have $\operatorname{tr}(A') = \operatorname{tr}(C^{-1}AC) = \operatorname{tr}(C^{-1}(AC)) = \operatorname{tr}((AC)C^{-1}) = \operatorname{tr}(A(CC^{-1})) = \operatorname{tr}(A)$. □

Definition 4.13. The <u>*trace*</u> of a linear operator $\mathcal{A}: V \to V$ is the trace of the matrix of $\mathcal{A}$ with respect to some (and hence any) basis of V.

Definition 4.14. <u>*Null space*</u> (<u>*kernel*</u>) of a linear operator $\mathcal{A}: V \to V$ is the set of all vectors $v \in V$ such that $\mathcal{A}\,v = 0$: $\operatorname{null}(\mathcal{A}) = \{v \in V \mid \mathcal{A}\,v = 0\}$.

Definition 4.15. <u>*Range*</u> (<u>*image*</u>) of a linear operator $\mathcal{A}: V \to V$ is the set of all vectors $w \in V$ such that $w = \mathcal{A}\,v$ for some $v \in V$: $\operatorname{range}(\mathcal{A}) = \{\mathcal{A}\,v \mid v \in V\}$.

Lemma 4.16. The null space and range of a linear operator are subspaces of V.

Exercise 4.19. Prove the lemma above.

Exercise 4.20. What is the relation between the range and the rank of a linear operator?

Remark 4.21. Note that we defined the null space and range of a linear operator without fixing a basis. The previous lemma can be proven without fixing a basis as well.

Now let us choose a basis in V and write down the matrix of the operator A. The null space is the subspace of solutions of the homogeneous system of linear equations $Ax = 0$. The range is the span of the columns of the matrix A. Since the sum of the dimensions of these two subspaces is equal to the dimension of V, we have the following theorem:

Theorem 4.17 (Rank-nullity theorem)**.** Let $\mathcal{A}: V \to V$ be a linear operator on a finite-dimensional vector space V. Then $\dim \operatorname{null}(\mathcal{A}) + \dim \operatorname{range}(\mathcal{A}) = \dim V$.

Definition 4.18. A linear operator $\mathcal{A}: V \to V$ is called <u>*nondegenerate*</u> if it is invertible, i.e., if there exists a linear operator $\mathcal{A}^{-1}: V \to V$ such that $\mathcal{A}\,\mathcal{A}^{-1} = \mathcal{A}^{-1}\mathcal{A} = I$.

Lemma 4.19 (Characterization of nondegenerate operators)**.** The following statements are equivalent for a linear operator $\mathcal{A}: V \to V$ on a finite-dimensional vector space V:

(1) $\mathcal{A}$ is nondegenerate;

(2) $\mathcal{A}$ is injective;

(3) $\operatorname{null}(\mathcal{A}) = \{0\}$;

(4) $\mathcal{A}$ is surjective;

(5) $\mathcal{A}$ is bijective;

(6) $\operatorname{rk}(\mathcal{A}) = \dim V$;

(7) $\det(\mathcal{A}) \neq 0$.

Exercise 4.22. Prove the lemma above. (Hint: use the rank-nullity theorem and review Linear algebra 1.)

Exercise 4.23. Find range, null space, rank, determinant, trace for differentiation of polynomials on $\mathbb{R}[x, 3]$.

4.4 Special examples of linear operators

Definition 4.20. A linear operator $A: V \to V$ is called *nilpotent* if there exists a positive integer k such that $A^k = 0$.

Exercise 4.24. Find the determinant of a nilpotent operator. What values can its rank take?

Definition 4.21. A linear operator $A: V \to V$ is called *idempotent* if $A^2 = A$.

Exercise 4.25. Find all nondegenerate idempotent operators. Provide an example of a nonzero degenerate idempotent operator.

Definition 4.22. A linear operator $\mathcal{P}: V \to V$ is called a projection if there exist two subspaces U and W of V such that $V = U \oplus W$ and $\mathcal{P}(u) = u$ for all $u \in U$ and $\mathcal{P}(w) = 0$ for all $w \in W$.

Exercise 4.26. Find the matrix of a projection in a basis consisting of bases of the subspaces U and W.

Exercise 4.27. Prove that a linear operator is a projection if and only if it is idempotent.

Then we can use the following equivalent definition of a projection:

Definition 4.23. A linear operator $\mathcal{P}: V \to V$ is called a projection if $\mathcal{P}^2 = \mathcal{P}$.

4.5 Dual space and dual operator

Definition 4.24. The *dual space* to a vector space V is the vector space

$$V^* = \{\, \varphi: V \to \mathbb{F} \mid \varphi \text{ is linear} \,\}.$$

Elements of V^* are called *linear functionals*.

Definition 4.25. Let $e = (e_1, \dots, e_n)$ be a basis of V. The *dual basis* $e^* = (e^1, \dots, e^n)$ in V^* is defined by

$$e^i(e_j) = \delta^i_j \quad (1 \le i, j \le n), \quad \text{where } \delta^i_j \text{ is the Kronecker delta: } \delta^i_j = 1 \text{ if } i = j \text{ and } \delta^i_j = 0 \text{ if } i \neq j.$$

Definition 4.26. Let $\mathcal{A}: V \to V$ be a linear operator. The *dual operator* is the linear operator

$$\mathcal{A}^*: V^* \to V^*, \qquad \mathcal{A}^*\varphi = \varphi \circ \mathcal{A}.$$

This means, that $(\mathcal{A}^*\varphi)(v) = \varphi(\mathcal{A}v)$.

Exercise 4.28. Check that the dual operator is indeed a *linear* operator.

Lemma 4.27. * [Matrix of the dual operator] Let $e = (e_1, \dots, e_n)$ be a basis of V and $e^* = (e^1, \dots, e^n)$ the dual basis of V^*. If A is the matrix of $\mathcal{A}$ in the basis e, then the matrix of $\mathcal{A}^*$ in the basis e^* is A^T (the transpose of A).

Proof. By definition, for every $\varphi \in V^*$ and $v \in V$ we have $(\mathcal{A}^*\varphi)(v) = \varphi(\mathcal{A}v)$. Writing both sides in coordinates with respect to e and e^* gives the equality of matrices described above. $\square$

Corollary 4.28. * For any linear operator $\mathcal{A}: V \to V$:

(1) $\det(\mathcal{A}^*) = \det(\mathcal{A})$;

(2) $\operatorname{tr}(\mathcal{A}^*) = \operatorname{tr}(\mathcal{A})$;

(3) $\operatorname{rk}(\mathcal{A}^*) = \operatorname{rk}(\mathcal{A})$.

Exercise 4.29. * Let $V = \mathbb{R}[x, 3]$ and let $\operatorname{ev}_a \in V^*$ be the evaluation functional $\operatorname{ev}_a(p) = p(a)$. Describe $\mathcal{A}^*$ explicitly for the differentiation operator $\mathcal{A} = \mathrm{D}$ and for the shift operator $(\mathcal{A}p)(x) = p(x+1)$.

4.6 Problems

Problem 4.1. Let A be a matrix of a linear operator $\mathcal{A}$ on a vector space V with respect to some basis e, and v be a vector, written in coordinates with respect to the same basis e. Find the coordinates of the image of v under the linear operator $\mathcal{A}$ in the same basis e:

a) $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$;

b) $A = \begin{pmatrix} 10 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$;

c) $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 100 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

Problem 4.2. Find the matrix of the operator:

a) $\varphi(e_1) = e_2$, $\varphi(e_2) = e_3$, $\varphi(e_3) = e_1$ and $\varphi(e_4) = e_5$, $\varphi(e_5) = e_4$ on $\mathbb{R}^5$ with respect to the basis $(e_1, e_2, e_3, e_4, e_5)$;

b) $(x_1, x_2, x_3) \mapsto (x_1,\ x_1 + 2x_2,\ x_2 + 3x_3)$ in $\mathbb{R}^3$ with respect to the standard basis;

c) rotation of three-dimensional space by angle $2\pi/3$ about the x-axis, with respect to the standard basis;

d) projection of $\mathbb{R}^3$ onto the e_2-axis along the coordinate plane spanned by e_1 and e_3, in the basis (e_1, e_2, e_3);

e) $X \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} X$ on the space $\mathbf{M}_2(\mathbb{R})$ with respect to the basis of matrix units;

f) $X \mapsto X \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ on the space $\mathbf{M}_2(\mathbb{R})$ with respect to the basis of matrix units;

g) $X \mapsto X^{\mathsf{T}}$ on the space $\mathbf{M}_2(\mathbb{R})$ with respect to the basis of matrix units;

h) $X \mapsto AXB$ (with fixed matrices A and B) on the space $\mathbf{M}_2(\mathbb{R})$ with respect to the basis of matrix units;

i) $X \mapsto AX + XB$ (with fixed matrices A and B) on the space $\mathbf{M}_2(\mathbb{R})$ with respect to the basis of matrix units;

j) differentiation on $\mathbb{R}[x]_n$ in the basis $(1, x, \ldots, x^n)$;

k) differentiation on $\mathbb{R}[x]_n$ in the basis $(x^n, x^{n-1}, \ldots, 1)$;

l) differentiation on $\mathbb{R}[x]_n$ in the basis $\left(1, x-1, \frac{(x-1)^2}{2}, \ldots, \frac{(x-1)^n}{n!}\right)$.

Problem 4.3. Find images, null spaces, ranks, determinants, traces for the operators from Problem 4.2.

Problem 4.4. Let a linear operator on $\mathbb{R}^3$ have, in the basis

$$((8,-6,7),(-16,7,-13),(9,-3,7)),$$

the matrix

$$\begin{pmatrix} 1 & -18 & 15 \\ -1 & -22 & 20 \\ 1 & -25 & 22 \end{pmatrix}.$$

Find its matrix in the basis

$$((1,-2,1),(3,-1,2),(2,1,2)).$$

Problem 4.5. Prove that in $\mathbb{R}^3$ there exists a unique linear operator that maps the vectors $(1,1,1)$, $(0,1,0)$, $(1,0,2)$ to the vectors $(1,1,1)$, $(0,1,0)$, $(1,0,1)$, respectively, and find the matrix of this operator in the standard basis (the basis of unit vectors).

Problem 4.6. Let a linear operator on a vector space V have, in the basis $(e_1,\ldots,e_4)$, the matrix

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 5 & 4 & 0 & -1 \\ 3 & 2 & 0 & 3 \\ 6 & 1 & -1 & 7 \end{pmatrix}.$$

Find the matrix of this operator with respect to the following bases:

a) (e_2,e_1,e_3,e_4);

b) $(e_1,e_1+e_2,e_1+e_2+e_3,e_1+e_2+e_3+e_4)$.

Problem 4.7. Let a linear operator on $\mathbb{R}_2[x]$ have, in the basis $(1,x,x^2)$, the matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Find its matrix in the basis

$$\left(3x^2+2x+1, x^2+3x+2, 2x^2+x+3\right)$$

Problem 4.8. Provide an example of a matrix A of size at least 2×2 such that $A\neq I$, $A^{100}=I$, and $A^n\neq I$ for all $0<n<100$.

Problem 4.9. Let a finite-dimensional vector space V be a direct sum of its subspaces: $V=U_1\oplus U_2\oplus U_3$. Let $\mathcal{B}$ be a linear operator on V such that for all $i=1,2,3$ we have $u\in U_i \Rightarrow \mathcal{B}(u)\in U_i$. Find the matrix of $\mathcal{B}$ in a basis of V that consists of bases of the subspaces U_1,U_2,U_3.

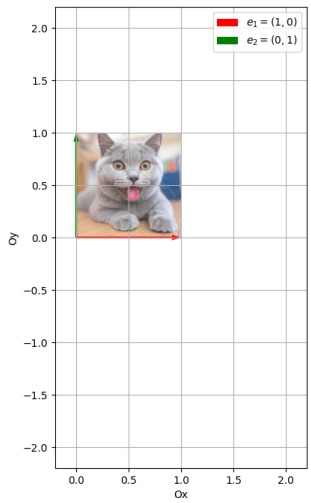
5 Eigenvalues and Where to Find Them

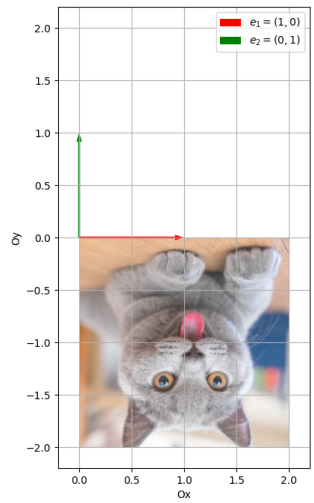
5.1 Motivation for eigenvalues and eigenvectors

Example 5.1. Question 1: What is the geometric meaning of the linear operator with matrix

$$A = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}?$$

See the figures:





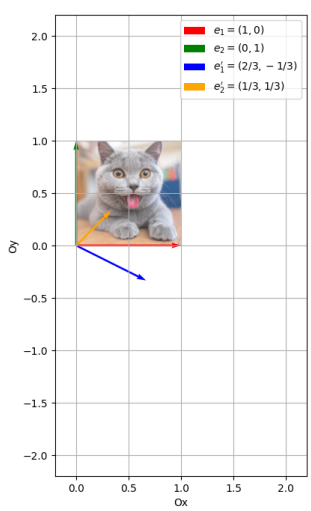
Question 2: What is the geometric meaning of the linear operator with matrix

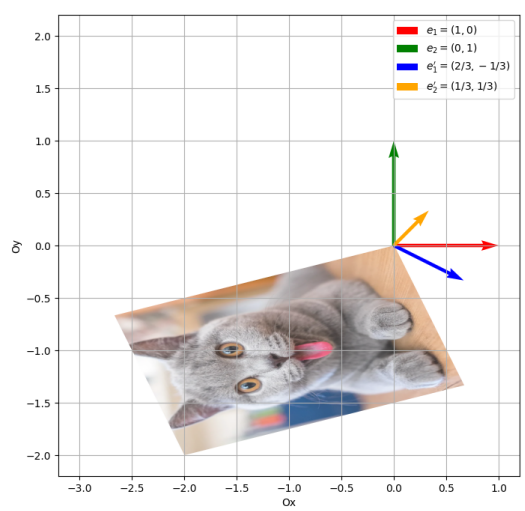
$$B = \begin{pmatrix} \frac{2}{3} & -\frac{8}{3} \\ -\frac{4}{3} & -\frac{2}{3} \end{pmatrix}?$$

It turns out that after a change of basis we obtain the following relation:

$$B' = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix}^{-1} \begin{pmatrix} \frac{2}{3} & -\frac{8}{3} \\ -\frac{4}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

This means that the second operator does the same thing, but relative to a different basis (see the figures).





If for any linear operator we could find a basis in which it simply stretches the basis vectors (or reflects and stretches), that would be ideal. Unfortunately, such a basis does not exist for every linear operator, but we will try to get as close to this as possible.

5.2 Definition of eigenvalues and eigenvectors

Definition 5.1. Let $T: V \to V$ be a linear operator on a vector space V over a field $\mathbb{F}$. A scalar $\lambda \in \mathbb{F}$ is called an *eigenvalue* of T if there exists a **nonzero** vector $v \in V$ such that $T(v) = \lambda v$. The vector v is called an *eigenvector* of T corresponding to the eigenvalue λ.

Remark 5.2. The zero vector is never considered an eigenvector, even though $T(\mathbf{0}) = \lambda \mathbf{0}$ for any λ.

Exercise 5.3. Find eigenvalues and eigenvectors in Example 5.1.

Definition 5.2. Let $\mathcal{A}$ be a linear operator on a finite-dimensional vector space V over a field $\mathbb{F}$. Then the polynomial $\chi_{\mathcal{A}}(t) = \det(tI - \mathcal{A}) \in \mathbb{F}[t]$ is called the *characteristic polynomial* of $\mathcal{A}$.

Since the determinant of a linear operator is basis-independent, the characteristic polynomial is well-defined and can be computed from any matrix of the operator (in any basis).

Example 5.4. Find the characteristic polynomial of the following operators:

1. zero operator on $\mathbb{R}^n$,
2. identity operator on $\mathbb{R}^n$,
3. operators from Example 5.1,
4. projection operator on $\mathbb{R}^n$.

If $\dim(V) = n$, then $\deg \chi_{\mathcal{A}}(t) = n$. Consider the coefficients of the characteristic polynomial. The leading coefficient is 1; the constant term is $\chi_{\mathcal{A}}(0) = \det(-\mathcal{A}) = (-1)^n \det(\mathcal{A})$.

Exercise 5.5. Find the coefficient of t^{n-1} in the characteristic polynomial of a linear operator.

Lemma 5.3. The eigenvalues of a linear operator are the roots of its characteristic polynomial.

Proof. Let λ be an eigenvalue of $\mathcal{A}$. Then there exists a nonzero vector $v \in V$ such that $\mathcal{A}(v) = \lambda v$. Then $(\lambda I - \mathcal{A})(v) = 0$. It means that $\lambda I - \mathcal{A}$ is a degenerate matrix, so $\det(\lambda I - \mathcal{A}) = 0$. Therefore, λ is a root of the characteristic polynomial.

If λ is a root of the characteristic polynomial, then $\det(\lambda I - \mathcal{A}) = 0$. It means that $\lambda I - \mathcal{A}$ is a degenerate matrix, so there exists a nonzero vector $v \in V$ such that $(\lambda I - \mathcal{A})(v) = 0$ (v is a nonzero solution of the homogeneous system of linear equations with coefficient matrix $\lambda I - \mathcal{A}$). Therefore, λ is an eigenvalue of $\mathcal{A}$. □

Remark 5.6. Note that the definition of an eigenvalue of a linear operator does not depend on the choice of basis. The same eigenvector may have different coordinates in different bases.

Exercise 5.7. Prove that if the matrix of an operator in some basis is upper triangular (or lower triangular, or diagonal), then its eigenvalues are exactly the entries on the main diagonal.

5.3 Existence of eigenvalues and eigenvectors

Lemma 5.4. Every linear operator on a complex finite-dimensional vector space has an eigenvalue and an eigenvector.

Proof. This follows immediately from the fundamental theorem of algebra. □

Example 5.8. *Why the complex field is necessary?* Because some polynomials have no roots. Consider a linear operator on $\mathbb{R}^2$ with the matrix

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The characteristic polynomial of this operator is $t^2 + 1$, which has no real roots. Therefore, this operator has no real eigenvalues and eigenvectors in $\mathbb{R}^2$.

Example 5.9. *Why the finite dimension is necessary?* Consider the linear operator $\varphi(p(x)) = xp(x)$ on $\mathbb{R}[x]$. Since for any non-zero polynomial $p(x)$, $\deg \varphi(p(x)) = \deg(p(x)) + 1$, then there are no nonzero polynomials, s. t. $\varphi(p(x)) = cp(x)$, where $c \in \mathbb{R}$.

It follows from basic polynomial theory that there are at most n eigenvalues. Over $\mathbb{C}$, counting multiplicities, there are exactly n eigenvalues because the equation $\det(tI - \mathcal{A}) = 0$ has exactly n complex roots.

5.4 Diagonalizable operators

If we have found at least one eigenvector, we can already simplify the matrix of a linear operator by zeroing the first column below the top entry.

What comes next? Any vector proportional to an eigenvector is also an eigenvector with the same eigenvalue, so we need to look for other eigenvectors. If we can find n linearly independent eigenvectors, we can form a basis and bring the matrix of the linear operator to diagonal form. To understand when this is guaranteed to be possible, we prove the following lemma.

Lemma 5.5. Eigenvectors corresponding to different eigenvalues are linearly independent.

Proof. Prove this by induction on k. One eigenvector is linearly independent, because it is nonzero. Assume that $k-1$ eigenvectors are linearly independent and consider k eigenvectors.

Let $\lambda_1, \lambda_2, \ldots, \lambda_k$ be different eigenvalues of $\mathcal{A}$ and $v_1, v_2, \ldots, v_k$ be corresponding eigenvectors. Then $\mathcal{A}(v_i) = \lambda_i v_i$ for $i = 1, 2, \ldots, k$. Consider the linear combination $c_1 v_1 + c_2 v_2 + \ldots + c_k v_k = 0$. Applying $\mathcal{A}$ to both sides, we get $c_1\lambda_1 v_1 + c_2\lambda_2 v_2 + \ldots + c_k\lambda_k v_k = 0$. Subtracting the first equation from the second, we get $c_2(\lambda_2 - \lambda_1)v_2 + \ldots + c_k(\lambda_k - \lambda_1)v_k = 0$. Since $\lambda_i \neq \lambda_j$ for $i \neq j$ and $k-1$ eigenvectors are linearly independent, we have $c_2 = 0, \ldots, c_k = 0$. Therefore, $c_1 = 0$ and the eigenvectors are linearly independent. $\square$

Corollary 5.6. If an operator has n different eigenvalues, then it is diagonalizable in some basis (for example, in the basis of eigenvectors).

Note that even if a linear operator has fewer than n distinct eigenvalues, if we happen to find n linearly independent eigenvectors then the operator is diagonalizable in the basis of its eigenvectors.

5.5 Problems

Problem 5.1. Find the eigenvalues and eigenvectors of the differentiation operator on $\mathbb{R}_n[x]$.

Problem 5.2. Find the eigenvalues and eigenvectors of the linear operators given, in some basis, by the following matrices:

a) $\begin{pmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$,

b) $\begin{pmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{pmatrix}$,

c) $\begin{pmatrix} 4 & -5 & 2 \\ 5 & -7 & 3 \\ 6 & -9 & 4 \end{pmatrix}$,

d) $\begin{pmatrix} 7 & -12 & 6 \\ 10 & -19 & 10 \\ 12 & -24 & 13 \end{pmatrix}$,

e) $\begin{pmatrix} 4 & -5 & 7 \\ 1 & -4 & 9 \\ -4 & 0 & 5 \end{pmatrix}$,

f) $\begin{pmatrix} 3 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 3 & 0 & 5 & -3 \\ 4 & -1 & 3 & -1 \end{pmatrix}$.

Problem 5.3. Find the eigenvalues and eigenvectors of the operators given, in some basis, by the following matrices. Can these operators be diagonalized? If yes, find a corresponding basis and write the change-of-basis matrix and the resulting diagonal matrix.

a) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

b) $\begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$,

c) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$,

d) $\begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$,

e) $\begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix}$,

f) $\begin{pmatrix} 4 & 7 & -5 \\ -4 & 5 & 0 \\ 1 & 9 & -4 \end{pmatrix}$,

g) $\begin{pmatrix} 4 & 2 & -5 \\ 6 & 4 & -9 \\ 5 & 3 & -7 \end{pmatrix}$,

h) $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$.

Problem 5.4. Prove and apply basic relations between eigenvalues, determinant, and trace:

a) Prove that for a 2×2 matrix, the product of its eigenvalues equals the determinant of the matrix, and their sum equals the trace.

b) Without computing the characteristic polynomial, find the eigenvalues of the matrix

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Problem 5.5. Compute A^{100} for each matrix:

a) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$,

b) $\begin{pmatrix} 7 & -4 \\ 14 & -8 \end{pmatrix}$,

c) $\begin{pmatrix} 1 & -3 & 4 \\ 4 & -7 & 8 \\ 6 & -7 & 7 \end{pmatrix}$,

d) $\begin{pmatrix} 4 & -5 & 7 \\ 1 & -4 & 9 \\ -4 & 0 & 5 \end{pmatrix}$,

e) $\begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$.

Problem 5.6. Find a closed formula for the n-th Fibonacci number. Recall that the Fibonacci numbers are defined by

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2.$$

Problem 5.7. Find A^n for any natural number n for the matrix

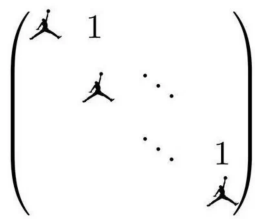
$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & -1 \\ 1 & 3 & -2 \end{pmatrix}.$$

Problem 5.8. Find a closed formula for the solution of the following recurrence relations:

a) $u(n) = 3u(n-1) - 2u(n-2), \quad u(0) = -2, \quad u(1) = 1;$

b) $u(n) = -2u(n-1) - u(n-2), \quad u(0) = -1, \quad u(1) = -1.$

6 Triangular and Jordan Normal Forms



In this section, we consider linear operators when a basis of eigenvectors is not available.

As usual, we consider a linear operator $\mathcal{A}: V \to V$, where V is a vector space over $\mathbb{F}$ and $\dim V = n$. In this section, we work over $\mathbb{C}$ (or over $\mathbb{R}$ when the characteristic polynomial has n real roots counted with multiplicity).

6.1 Invariant subspaces

Definition 6.1. The eigenspace of a linear operator $\mathcal{A}: V \to V$ corresponding to an eigenvalue λ is the set of vectors $V_\lambda(\mathcal{A}) = \{v \in V \mid \mathcal{A}(v) = \lambda v\} = \operatorname{null}(\mathcal{A} - \lambda I)$.

Exercise 6.1. The eigenspace is a subspace of V.

Remark 6.2. The set of all eigenvectors corresponding to a given eigenvalue λ is $V_\lambda(\mathcal{A}) \setminus \{0\}$.

Definition 6.2. A subspace W of V is called invariant under a linear operator $\mathcal{A}: V \to V$ if $\mathcal{A}(W) \subseteq W$.

Exercise 6.3. If a linear operator has an eigenvector, then the subspace spanned by that eigenvector is invariant under the linear operator.

Exercise 6.4. If a linear operator has an eigenspace, then that eigenspace is invariant under the linear operator.

Exercise 6.5. If a subspace W is invariant under a linear operator $\mathcal{A}: V \to V$, then the restriction of $\mathcal{A}$ to W is a linear operator on W.

Exercise 6.6. What is the form of the matrix of a linear operator in a basis adapted to an invariant subspace? (The invariant subspace's basis is a part of the basis of the whole space.)

Exercise 6.7. What happens to the matrix if the space is represented as a direct sum of invariant subspaces?

Definition 6.3. The algebraic multiplicity of an eigenvalue λ is the multiplicity of λ as a root of the characteristic polynomial.

Definition 6.4. The geometric multiplicity of an eigenvalue λ is the dimension of the eigenspace $V_\lambda(\mathcal{A})$.

Lemma 6.5. The algebraic multiplicity of an eigenvalue λ is greater than or equal to its geometric multiplicity.

Proof. The eigenspace $V_\lambda(\mathcal{A})$ is an invariant subspace of $\mathcal{A}$. Extend it to a basis of the whole space and write the matrix of $\mathcal{A}$ in this basis:

$$A = \begin{pmatrix} \lambda I_k & * \\ 0 & B \end{pmatrix},$$

where $k = \dim V_\lambda(\mathcal{A})$.

Then the characteristic polynomial of $\mathcal{A}$ equals $(t-\lambda)^k \det(tI_{n-k} - B)$. Hence, the algebraic multiplicity of the eigenvalue λ is greater than or equal to $k = \dim V_\lambda(\mathcal{A})$. □

Lemma 6.6. The sum of the eigenspaces of a linear operator corresponding to different eigenvalues is direct.

Exercise 6.8. Prove the lemma.

Then the following theorem holds:

Theorem 6.7. A linear operator is diagonalizable if and only if the algebraic multiplicity of each eigenvalue is equal to its geometric multiplicity.

Exercise 6.9. Prove the theorem.

We will now spend some time understanding what to do when an operator is not diagonalizable.

6.2 Triangular Form

First, we show that any complex linear operator can be brought to an upper-triangular form in some basis.

From the previous section, we know that every complex linear operator has at least one one-dimensional invariant subspace (spanned by an eigenvector).

Lemma 6.8. Any complex linear operator has an $(n-1)$-dimensional invariant subspace.

Proof. Consider the linear operator $\mathcal{A}^*: V^* \to V^*$ over $\mathbb{C}$ with $\dim V = \dim V^* = n$. Then $\mathcal{A}^*$ has an eigenvector $f \in V^*$ with eigenvalue $\lambda \in \mathbb{C}$. Since f is a nonzero linear functional (a single row in coordinates), its kernel (the solution set of the homogeneous linear equation $f(v) = 0$) has dimension $n-1$.

Let us prove that $\ker f$ is invariant under $\mathcal{A}$. Assume $v \in \ker f$. Then $0 = f(v) = \lambda f(v) = (\lambda f)(v) = (\mathcal{A}^* f)(v) = f(\mathcal{A} v)$, hence $\mathcal{A} v \in \ker f$. Thus, $\ker f$ is an $(n-1)$-dimensional invariant subspace of $\mathcal{A}$. □

Corollary 6.9. Any complex linear operator has an upper-triangular matrix in some basis.

Exercise 6.10. Prove the corollary by induction on the dimension of the space.

6.3 Annihilating polynomials

We now work with polynomials in matrices and operators. For a polynomial $p(x) = a_m x^m + \cdots + a_1 x + a_0$, define the polynomial in a matrix A as $p(A) = a_m A^m + \cdots + a_1 A + a_0 I$ (I is the identity matrix of the same size as A) and the polynomial in an operator $\mathcal{A}$ as $p(\mathcal{A}) = a_m \mathcal{A}^m + \cdots + a_1 \mathcal{A} + a_0 I$ (I is the identity on V).

Exercise 6.11. Prove that $p(\mathcal{A})$ is a linear operator on V.

Exercise 6.12. Let $p(x) = x^2 - 3x + 1$. Compute $p(A)$ for

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Exercise 6.13. Show that for any polynomials $p(x)$ and $q(x)$, we have $p(A)q(A) = q(A)p(A)$ (even though two arbitrary matrices or operators do not commute).

Definition 6.10. An annihilating polynomial of a linear operator $\mathcal{A}$ is a polynomial $p(x)$ such that $p(\mathcal{A}) = 0$ (the zero operator on the right-hand side).

Exercise 6.14. Prove that $x^2 - 1$ is an annihilating polynomial of the matrix

$$A = \begin{pmatrix} 1 & -4 \\ 0 & -1 \end{pmatrix}.$$

Exercise 6.15. Prove that x^3 is an annihilating polynomial of the matrix

$$A = \begin{pmatrix} 0 & 2 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

(Recall the definition of a nilpotent matrix from the previous section.)

Exercise 6.16. Find an annihilating polynomial for a projection (idempotent) matrix.

Definition 6.11. Minimal polynomial of a linear operator $\mathcal{A}$ is a monic polynomial $m(x)$ of the smallest degree such that $m(\mathcal{A}) = 0$.

Theorem 6.12. The minimal polynomial of a linear operator exists and is unique. Moreover, the degree of the minimal polynomial is less than or equal to n^2. (We will be able to even decrease it to n later.)

Proof. Consider elements

$$I, A, A^2, \ldots, A^{n^2}$$

in the n^2-dimensional vector space $M_n(\mathbb{F})$. Since there are $n^2 + 1$ elements, they are linearly dependent. Hence, there exist scalars $c_0, c_1, \ldots, c_{n^2}$, not all zero, such that

$$c_0 I + c_1 A + c_2 A^2 + \ldots + c_{n^2} A^{n^2} = 0.$$

Hence, $m(x) = c_0 + c_1 x + c_2 x^2 + \ldots + c_{n^2} x^{n^2}$ is an annihilating polynomial of $\mathcal{A}$.

Then we can choose an annihilating polynomial of the smallest degree. Dividing by the leading coefficient, we get a monic polynomial.

Now assume that there are two minimal polynomials $m(x)$ and $m'(x)$ of the same degree. Then $m(x) - m'(x)$ is an annihilating polynomial of degree less than the degree of $m(x)$ and $m'(x)$. Hence, $m(x) = m'(x)$. □

Theorem 6.13. Any annihilating polynomial of a linear operator $\mathcal{A}$ is divisible by the minimal polynomial of the operator.

Exercise 6.17. Prove the theorem using the division algorithm for polynomials.

Exercise 6.18. Find the minimal polynomial of a projection, a nilpotent operator, and a rotation of the real plane by $2\pi/3$.

Theorem 6.14 (Cayley-Hamilton theorem)**.** The characteristic polynomial of a linear operator (and its matrix) is an annihilating polynomial of the operator (and the matrix).

Proof. Since this statement does not depend on the choice of basis, it is natural to apply Corollary 6.9 and assume that the matrix A in some basis $(e_1, \ldots, e_n)$ is upper triangular. Consider the chain of $\mathcal{A}$-invariant subspaces

$$V = V_0 \supset V_1 \supset \ldots \supset V_{n-1} \supset V_n = 0$$

where $V_k = \langle e_1, e_2, \ldots, e_{n-k} \rangle$. Since $(\mathcal{A} - \lambda_{n-k} I)\, e_{n-k} \in V_{k+1}$, it follows that

$$(\mathcal{A} - \lambda_{n-k} I)\, V_k \subset V_{k+1}.$$

Therefore,

$$\begin{aligned} \chi_{\mathcal{A}}(\mathcal{A}) V &= \prod_{i=1}^{n} (\mathcal{A} - \lambda_i I)\, V = \\ &= (\mathcal{A} - \lambda_1 I) \ldots (\mathcal{A} - \lambda_n I)\, V_0 \subset (\mathcal{A} - \lambda_1 I) \ldots (\mathcal{A} - \lambda_{n-1} I)\, V_1 \subset \\ &\subset (\mathcal{A} - \lambda_1 I) \ldots (\mathcal{A} - \lambda_{n-2} I)\, V_2 \subset \ldots \subset (\mathcal{A} - \lambda_1 I)\, V_{n-1} = 0 \end{aligned}$$

Consequently, $\chi_{\mathcal{A}}(\mathcal{A}) V = 0$, that is, $\chi_{\mathcal{A}}(\mathcal{A}) = 0$. □

6.4 Generalized eigenspaces

Definition 6.15. *Generalized eigenspace* of a linear operator $\mathcal{A} : V \to V$ corresponding to an eigenvalue λ is the set of vectors $V^\lambda = \{v \in V \mid (\mathcal{A} - \lambda I)^k v = 0 \text{ for some } k \geqslant 0\}$.

Exercise 6.19. Prove that the generalized eigenspace is a subspace of V that is invariant under $\mathcal{A}$.

Definition 6.16. *Generalized eigenvector of rank* m corresponding to the eigenvalue λ is a vector v such that $(\mathcal{A} - \lambda I)^m v = 0$ and $(\mathcal{A} - \lambda I)^{m-1} v \neq 0$.

The unique generalized eigenvector of rank 0 is the zero vector; generalized eigenvectors of rank 1 are the ordinary eigenvectors.

Consider the following sequence of subspaces:

$$\{0\} \subset \operatorname{null}(\mathcal{A} - \lambda I) \subset \operatorname{null}(\mathcal{A} - \lambda I)^2 \subset \ldots \subset \operatorname{null}(\mathcal{A} - \lambda I)^m \subset \ldots.$$

It cannot be strictly increasing infinitely, because the dimension of the space is finite. Moreover, if for the first time at some m we have $\operatorname{null}(\mathcal{A} - \lambda I)^m = \operatorname{null}(\mathcal{A} - \lambda I)^{m+1}$, then the chain stabilizes: all later subspaces $\operatorname{null}(\mathcal{A} - \lambda I)^k$ $(k > m)$ are equal to $\operatorname{null}(\mathcal{A} - \lambda I)^m$. In this case $m \leqslant \dim V$. Hence, $V^\lambda = \{v \in V \mid (\mathcal{A} - \lambda I)^n v = 0\}$, where $n = \dim V$.

We have seen earlier that for some operators we cannot find a basis of eigenvectors (even over $\mathbb{C}$). However, we can find a basis of generalized eigenvectors. Our next goal is to prove that for any linear operator on a finite-dimensional complex vector space V, the space V can be represented as a direct sum of generalized eigenspaces corresponding to different eigenvalues.

Theorem 6.17. Let $\mathcal{A} : V \to V$ be a linear operator, $\dim V = n$. And let its characteristic polynomial have n roots counting multiplicities:

$$\chi_{\mathcal{A}}(t) = \prod_{i=1}^{p} (t - \lambda_i)^{n_i}\,; \quad \lambda_i \neq \lambda_j \text{ for } i \neq j$$

(always true over $\mathbb{C}$).

Then

(1) $V = V^{\lambda_1} \oplus \ldots \oplus V^{\lambda_p}$ is a direct sum of generalized eigenspaces corresponding to distinct eigenvalues;

(2) each generalized eigenspace is invariant under $\mathcal{A}$;

(3) $\dim V^{\lambda_i} = n_i$ (the algebraic multiplicity of λ_i) for each $i = 1, \ldots, p$;

(4) the operator $\mathcal{A} - \lambda_i I$ is nilpotent on V^{λ_i};

(5) the operator $\mathcal{A} - \lambda_i I$ is injective on

$$V_i = V(\lambda_1) \oplus \ldots \oplus V(\lambda_{i-1}) \oplus V(\lambda_{i+1}) \oplus \ldots \oplus V(\lambda_p)\,;$$

(6) λ_i is the only eigenvalue of the operator $\mathcal{A}|_{V^{\lambda_i}}$.

Proof. Introduce notation:

$$\chi_i(t) = \prod_{j \neq i} (t - \lambda_j)^{n_j}\,, \quad i = 1, 2, \ldots, p.$$

Since $(\chi_1(t), \ldots, \chi_p(t)) = 1$, there exist polynomials $f_1(t), \ldots, f_p(t)$ such that

$$\sum_{i=1}^{p} \chi_i(t) f_i(t) = 1.$$

Also define the subspaces

$$W_i = \chi_i(\mathcal{A}) f_i(\mathcal{A}) V = \{\chi_i(\mathcal{A}) f_i(\mathcal{A}) v \mid v \in V\}\,, \quad 1 \leqslant i \leqslant p.$$

The subspaces W_i

- are invariant under $\mathcal{A}$:

$$\mathcal{A} W_i = \mathcal{A}\, \chi_i(\mathcal{A}) f_i(\mathcal{A}) V = \chi_i(\mathcal{A}) f_i(\mathcal{A}) \mathcal{A} V \subset \chi_i(\mathcal{A}) f_i(\mathcal{A}) V = W_i;$$

- are contained in the corresponding subspaces V^{λ_i} (in fact, they coincide; we will see this later):

$$(\mathcal{A} - \lambda_i I)^{n_i} W_i = \chi_{\mathcal{A}}(\mathcal{A}) f_i(\mathcal{A}) V = [\text{by the Cayley–Hamilton theorem, } \chi_{\mathcal{A}}(\mathcal{A}) = 0] = 0,$$

  hence $W_i \subset V^{\lambda_i}$;

- their sum equals the whole space V: substitute the operator $\mathcal{A}$ into the identity decomposition and obtain

$$\sum_{i=1}^{p} \chi_i(\mathcal{A}) f_i(\mathcal{A}) = I,$$

  then any vector $v \in V$ can be written as

$$v = \sum_{i=1}^{p} \chi_i(\mathcal{A}) f_i(\mathcal{A}) v \in \sum_{i=1}^{p} W_i, \text{ hence } V = \sum_{i=1}^{p} W_i.$$

Therefore, from $W_i \subset V^{\lambda_i}$ we get

$$V = \sum_{i=1}^{p} V^{\lambda_i}.$$

Now we show that the sum of the generalized eigenspaces is direct. Let $v \in V^{\lambda_i} \cap V_i$, where $V_i = \sum_{j \neq i} V^{\lambda_j}$ (for now just a sum, not necessarily direct). Then $(\mathcal{A} - \lambda_i I)^n v = 0$, and since $v = \sum_{j \neq i} v_j$ with $(\mathcal{A} - \lambda_j I)^n v_j = 0$, we obtain

$$\left(\prod_{j \neq i} (\mathcal{A} - \lambda_j I)^n \right) v = 0.$$

Since the polynomials $(t - \lambda_i)^n$ and $c(t) = \prod_{j \neq i} (t - \lambda_j)^n$ are coprime, there exist polynomials $a(t), b(t)$ such that

$$a(t)\,(t - \lambda_i)^n + b(t)c(t) = 1.$$

It follows that

$$v = a(\mathcal{A})\,(\mathcal{A} - \lambda_i I)^n v + b(\mathcal{A}) \left(\prod_{j \neq i} (\mathcal{A} - \lambda_j I)^n \right) v = 0,$$

i.e., the spaces $V(\lambda_i)$ and V_i intersect trivially. Hence, the sum is indeed direct.

Now we see that $W_i = V^{\lambda_i}$ for all $i = 1, \ldots, p$. (1) is proved.

Since W_i are invariant under $\mathcal{A}$ and coincide with V^{λ_i}, the subspaces V^{λ_i} are invariant under $\mathcal{A}$. (2) is proved.

Next, since

$$(\mathcal{A} - \lambda_i I)^n V^{\lambda_i} = \mathbf{0}$$

(this was known even before the theorem), we obtain (4). Moreover, the minimal polynomial of $\mathcal{A}$ on V^{λ_i} divides $(t - \lambda_i)^n$ and even $(t - \lambda_i)^{n_i}$ (because it divides the characteristic polynomial of the whole operator). This immediately gives (6). Next, write the matrix of $\mathcal{A}$ in a basis obtained by concatenating bases of the direct summands: it has a block-diagonal form with blocks $A_1, \ldots, A_p$ on the diagonal, where A_i is the matrix of the restriction of $\mathcal{A}$ to V^{λ_i} of size $n_i' \times n_i'$. The characteristic polynomial of each block is then $(t - \lambda_i)^{n_i'}$ with $n_i' \leqslant n_i$. Since $n_1' + \ldots + n_p' = n$, we conclude that $n_i' = n_i$ for all $i = 1, \ldots, p$. (3) is proved.

It remains to prove (6), i.e. the nondegeneracy of the restriction $(\mathcal{A} - \lambda_i I)|_{V_i}$. Indeed, otherwise

$$(\operatorname{null}(\mathcal{A} - \lambda_i I)) \cap V_i \neq \{0\}$$

and $\mathcal{A}v - \lambda_i v = 0$ for some $0 \neq v \in V_i$. However, on V_i the characteristic polynomial of $\mathcal{A}$ is $\chi_i(t) = \prod_{j \neq i} (t - \lambda_j)^{n_j}$, so λ_i cannot be an eigenvalue. $\square$

6.5 Nilpotent Operators

For a nilpotent operator $\mathcal{A}$, let v be a generalized eigenvector of rank k (i.e., $\mathcal{A}^k v = 0$ but $\mathcal{A}^{k-1} v \neq 0$). Then $v, \mathcal{A}v, \ldots, \mathcal{A}^{k-1}v$ are linearly independent (proof by induction on k).

The cyclic subspace generated by v is the smallest invariant subspace containing v. It is convenient to write the chain backwards:

$$v = e_m, (\mathcal{A} - \lambda I)e_m = e_{m-1}, \ldots, (\mathcal{A} - \lambda I)^{m-1} v = (\mathcal{A} - \lambda I)e_2 = e_1, (\mathcal{A} - \lambda I)^m v = (\mathcal{A} - \lambda I)e_1 = 0.$$

Lemma 6.18. If v is a generalized eigenvector of rank k, then the vectors $v, (\mathcal{A} - \lambda I)v, \ldots, (\mathcal{A} - \lambda I)^{k-1} v$ are linearly independent.

It follows that the cyclic subspace has dimension k, and on it the matrix of $\mathcal{A}$ has the Jordan form $J_k(\lambda)$:

$$\begin{pmatrix} \lambda & 1 & 0 & \ldots & 0 \\ 0 & \lambda & 1 & \ldots & 0 \\ 0 & 0 & \lambda & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & \lambda \end{pmatrix}$$

in the basis $v, (\mathcal{A} - \lambda I)v, \ldots, (\mathcal{A} - \lambda I)^{k-1} v$.

Lemma 6.19. Every nilpotent operator admits a decomposition of the space into a direct sum of cyclic subspaces.

Corollary 6.20. Any nilpotent operator can be brought to a block-diagonal form with Jordan blocks with eigenvalue 0 on the diagonal.

Therefore, $\mathcal{A} - \lambda I$ is in Jordan normal form on its generalized eigenspace. Adding back λI gives the Jordan form for $\mathcal{A}$ on the generalized eigenspace; performing this for all generalized eigenspaces yields the Jordan normal form of $\mathcal{A}$.

Theorem 6.21. The sizes and numbers of Jordan blocks are uniquely determined, i. e. the Jordan normal form is unique up to the ordering of the blocks.

6.6 How to compute the Jordan normal form

Step 1. Find the eigenvalues and their algebraic multiplicities.

Step 2. Find the eigenvectors and the geometric multiplicities of the eigenvalues. If the algebraic multiplicities equal the geometric multiplicities, write the diagonal Jordan normal form and, using the eigenvector basis, construct the change-of-basis matrix.

Step 3. If the operator is not diagonalizable, determine the sizes of the Jordan blocks. If the sizes are not obvious, compute the numbers $N(m, \lambda)$ of Jordan blocks of size m with eigenvalue λ using the formula:

$$N(m, \lambda) = r_{m-1} - 2r_m + r_{m+1},$$
$$m \geqslant 1, \quad r_t = \operatorname{rk}(\mathcal{A} - \lambda I)^t, \quad r_0 = n.$$

Step 4. Choose linearly independent generalized eigenvectors of the required ranks and build the corresponding cyclic subspaces. To do this, first solve $(A-\lambda I)^m v = 0$, where m is the first index at which the chain $\operatorname{null}(\mathcal{A} - \lambda I)^m$ stabilizes (if unknown, taking m equal to the algebraic multiplicity of λ suffices). Then pick any nonzero vector from this nullspace and build its cyclic subspace. Verify that the chain length equals the size of the corresponding Jordan block and that the last nonzero vector is an eigenvector (up to scaling). If the chain is too short, choose a different vector.

Step 5. Write the Jordan normal form and the change-of-basis matrix. Use the eigenvectors from the previous step for the new basis.

6.7 Problems

Problem 6.1. Find the Jordan normal form of matrices from problems 5.1, 5.2, and 5.3.

Problem 6.2. Find the Jordan normal form and transition matrix for each matrix:

(a) $\begin{pmatrix} 3 & 2 & -3 \\ 4 & 10 & -12 \\ 3 & 6 & -7 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 1 & -1 \\ -3 & -3 & 3 \\ -2 & -2 & 2 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 1 & -1 & 1 \\ -1 & 2 & -1 & 1 \\ -1 & 1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 6 & -9 & 5 & 4 \\ 7 & -13 & 8 & 7 \\ 8 & -17 & 11 & 8 \\ 1 & -2 & 1 & 3 \end{pmatrix}$

Problem 6.3. (a) Prove the formula for any natural n:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

(b) Find A^n for any natural n, where

$$A = \begin{pmatrix} -1 & 4 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 0 & 18 & 25 \\ 0 & 0 & 13 & -18 \end{pmatrix}.$$

7 Symmetric matrices

See Gilbert Strang, Introduction to Linear Algebra, Chapter 6.4, 6.5.

7.1 Preliminaries

Dot product of two vectors u, v (column vectors): $(u, v) = u^{\mathrm{T}} v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$.

Vectors are orthogonal if $(u, v) = 0$.

A basis is orthogonal if its vectors are pairwise orthogonal.

Norm of a real vector u: $\|u\| = \sqrt{(u, u)} = \sqrt{u^{\mathrm{T}} u} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$.

Norm of a complex vector u: $\|u\| = \sqrt{(u, u)} = \sqrt{\overline{u}^{\mathrm{T}} u} = \sqrt{|u_1|^2 + |u_2|^2 + \cdots + |u_n|^2}$.

A basis is orthonormal if its vectors are pairwise orthogonal and have norm 1.

7.2 Symmetric matrices

Definition 7.1. A matrix S is called symmetric if $S = S^{\mathrm{T}}$.

Example 7.1. The matrix

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

is symmetric. Let's find its eigenvalues and eigenvectors. We will see that all its eigenvalues are real and its eigenvectors are orthogonal.

Lemma 7.2. Let $S \in M_n(\mathbb{R})$ be a symmetric matrix ($S^{\mathrm{T}} = S$). Then it has n real eigenvalues (counted with multiplicity).

Lemma 7.3. Let $S \in M_n(\mathbb{R})$ be a symmetric matrix. Then it is diagonalizable (over $\mathbb{R}$).

Lemma 7.4. Let $S \in M_n(\mathbb{R})$ be a symmetric matrix. Then eigenvectors corresponding to different eigenvalues are orthogonal.

Lemma 7.5. Let $S \in M_n(\mathbb{R})$ be a symmetric matrix. Then it can be diagonalized by an orthonormal basis of eigenvectors. That is, there exists an orthogonal matrix C and a diagonal matrix D such that $S = CDC^{\mathrm{T}}$, where C is an orthogonal matrix: $C^{\mathrm{T}} = C^{-1}$.

Note that the columns of C form an orthonormal basis.

7.3 *Bilinear forms

Let $\mathbb{F}$ be a field of real or complex numbers. And let V be a vector space over $\mathbb{F}$.

Definition 7.6. A bilinear form is a function $\varphi : V \times V \to \mathbb{F}$ that is linear in each argument.

If we fix a basis $e_1, \ldots, e_n$ in V, then we can represent the bilinear form as a matrix A in this basis:

$$\varphi(u, v) = u^{\mathrm{T}} A v,$$

where $a_{ij} = \varphi(e_i, e_j)$.

Example 7.2. The dot product is a bilinear form. Find its matrix in the basis $e_1, e_2, \ldots, e_n$.

Lemma 7.7. Let A be a matrix of a bilinear form $\varphi : V \times V \to \mathbb{F}$ in the basis $e_1, \ldots, e_n$ and let C be a transition matrix from this basis to the basis $e'_1, \ldots, e'_n$. Then the matrix of the bilinear form φ in the basis $e'_1, \ldots, e'_n$ is $A' = C^{\mathrm{T}} A C$.

Definition 7.8. A bilinear form $\varphi : V \times V \to \mathbb{F}$ is called symmetric if $\varphi(u, v) = \varphi(v, u)$ for all $u, v \in V$.

Lemma 7.9. Let $\varphi : V \times V \to \mathbb{F}$ be a bilinear form. Then it is symmetric if and only if its matrix is symmetric (in any basis).

7.4 Positive definite matrices

Definition 7.10. A matrix $S \in M_n(\mathbb{R})$ is called positive definite if $x^{\mathrm{T}} S x > 0$ for all $x \neq 0$.

Definition 7.11. A matrix $A \in M_n(\mathbb{R})$ is called positive semidefinite if $x^{\mathrm{T}} A x \geqslant 0$ for all $x \in \mathbb{R}^n$.

Lemma 7.12. A symmetric matrix $S \in M_n(\mathbb{R})$ is positive definite if and only if all its eigenvalues are positive.

Lemma 7.13. A symmetric real matrix $S \in M_n(\mathbb{R})$ is positive definite if and only if there exists a real matrix $A \in M_{k,n}(\mathbb{R})$ with linearly independent columns such that $S = A^{\mathrm{T}} A$.

Lemma 7.14 (Sylvester's criterion)**.** A symmetric real matrix S is positive definite if and only if all its leading principal minors

$$\Delta_1 = a_{11}, \Delta_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \ldots, \Delta_i = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1i} \\ a_{21} & a_{22} & \ldots & a_{2i} \\ \ldots & \ldots & \ldots & \ldots \\ a_{i1} & a_{i2} & \ldots & a_{ii} \end{vmatrix}, \ldots$$

are positive.

7.5 Problems

Problem 7.1. Find the eigenvalues and the unit eigenvectors of the matrix

$$S = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

Find an orthogonal matrix C that diagonalizes S.

Problem 7.2. Find the diagonal form and an orthogonal change-of-basis matrix for the matrix

$$S = \begin{pmatrix} -2 & 6 \\ 6 & 7 \end{pmatrix}.$$

Problem 7.3. Find an orthogonal matrix Q that diagonalizes the matrix

$$S = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{pmatrix}.$$

Problem 7.4. Find all orthogonal matrices that diagonalize

$$S = \begin{pmatrix} 9 & 12 \\ 12 & 16 \end{pmatrix}.$$

Problem 7.5. Consider a real matrix

$$\begin{pmatrix} 1 & b \\ b & 1 \end{pmatrix}.$$

Find signs of its eigenvalues for all possible values of b.

Problem 7.6. Which of the following matrices are positive definite?

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ -2 & 6 \end{pmatrix}, \begin{pmatrix} -1 & 2 \\ 2 & -6 \end{pmatrix}, \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

For positive definite matrices, find its decomposition $S = A^T A$.

Problem 7.7. Which of S_1, S_2, S_3, S_4 have two positive eigenvalues? Also find a vector x such that $x^{\mathrm{T}} S_1 x < 0$, thus S_1 is not positive definite.

$$S_1 = \begin{pmatrix} 5 & 6 \\ 6 & 7 \end{pmatrix}, \quad S_2 = \begin{pmatrix} -1 & -2 \\ -2 & -5 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 10 \\ 10 & 100 \end{pmatrix} \quad S_4 = \begin{pmatrix} 1 & 10 \\ 10 & 101 \end{pmatrix}.$$

Problem 7.8. For which numbers b and c are these matrices positive definite?

$$S_1 = \begin{pmatrix} 1 & b \\ b & 9 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 2 & 4 \\ 4 & c \end{pmatrix}, \quad S_3 = \begin{pmatrix} c & b \\ b & c \end{pmatrix}.$$

Problem 7.9. Test whether $A^{\mathrm{T}}A$ is positive definite in each case; this requires A to have linearly independent columns.

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \quad \text{and} \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad \text{and} \quad A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}.$$

8 Singular Value Decomposition

8.1 Motivation for SVD

If A is an $m \times n$ matrix of rank r, then it can be written as

$$A = BC,$$

where B is an $m \times r$ matrix and C is an $r \times n$ matrix. Then instead of storing the whole matrix A (mn memory cells), we can store the matrices B and C in $(m+n)r$ memory cells.

Exercise 8.1. Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ -1 & -2 & 1 \end{pmatrix}.$$

Represent A as BC, where B is 3×1 matrix and C is 1×3 matrix.

Exercise 8.2. Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \\ 4 & 5 & -2 \end{pmatrix}.$$

Represent A as BC, where B is 3×2 matrix and C is 2×3 matrix.

We will construct a representation $A = BDC$, where D is a diagonal matrix.

Often the matrix A has full rank; nevertheless, for compression and noise reduction we will later discard the smallest diagonal entries of D (corresponding to the smallest singular values).

8.2 Definition of SVD

Lemma 8.1. Matrices AA^{T} and $A^{\mathrm{T}}A$ are symmetric and share the same non-zero eigenvalues.

Then there exist orthogonal matrices U and V and a diagonal matrix with nonnegative entries Λ such that

$$AA^{\mathrm{T}} = U\Lambda U^{\mathrm{T}},$$

$$A^{\mathrm{T}}A = V\Lambda V^{\mathrm{T}}$$

.

When A is a real matrix, the symmetric matrices AA^{T} and $A^{\mathrm{T}}A$ have real eigenvalues. If A has independent rows, then AA^{T} is positive definite and all its eigenvalues are positive. If it has independent columns, then $A^{\mathrm{T}}A$ is positive definite and all its eigenvalues are positive. In the general case, AA^{T} and $A^{\mathrm{T}}A$ are positive semidefinite and their eigenvalues are nonnegative.

Let us arrange the eigenvalues of AA^{T} (and $A^{\mathrm{T}}A$) in decreasing order:

$$\sigma_1^2 \geqslant \sigma_2^2 \geqslant \cdots \geqslant \sigma_r^2 > 0 = \sigma_{r+1}^2 = \cdots = \sigma_n^2,$$

where r is the rank of AA^T (and A^TA).

Denote a matrix $\Sigma = \operatorname{diag}(\sigma_1, \sigma_2, \cdots, \sigma_r, 0, \cdots, 0)$.

Then for orthogonal matrices U (eigenvectors of AA^{T}) and V (eigenvectors of $A^{\mathrm{T}}A$) we can write

$$AA^T = U\Sigma^2 U^{\mathrm{T}} = U\Sigma V^T V\Sigma U^T = (U\Sigma V^T)(U\Sigma V^T)^{\mathrm{T}}.$$

We call $A = U\Sigma V^{\mathrm{T}}$ the Singular Value Decomposition (SVD) of A. The numbers $\sigma_1, \sigma_2, \ldots, \sigma_{\min(m,n)}$ are the singular values of A, and the columns of U and V are the left and right singular vectors of A, respectively. The number of nonzero singular values equals the rank of A.

(We skip the rigorous proof of the existence of SVD and move to the algorithm using the observations above.)

8.3 Low-rank approximation

Definition 8.2. The Frobenius norm of a matrix A is defined as

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2}.$$

Theorem 8.3. Fix a matrix A and consider a problem

$$\begin{cases} \|A-B\|_F \to \min_B \\ \operatorname{rk} B \leqslant k \end{cases}$$

Then the solution of this problem is the matrix $B = U\Sigma_k V^{\mathrm{T}}$, where Σ_k is obtained from Σ by keeping only the first k singular values (and zeroing out the rest).

8.4 Pseudoinverse

Exercise 8.3. Let A be a nondegenerate matrix with known singular values $\sigma_1, \sigma_2, \cdots, \sigma_n$. Find the singular values of A^{-1}.

Definition 8.4. The pseudoinverse of a matrix A is defined as

$$A^+ = V\Sigma^+ U^{\mathrm{T}},$$

where Σ^+ is the matrix with the reciprocals of the non-zero singular values of A on the diagonal.

Note that if A is invertible, then $A^+ = A^{-1}$.

Consider the following problem: solve the matrix equation $Ax = b$. If A is invertible, then there is a unique solution $x = A^{-1}b$. If A is not invertible, then there can be more than one solution or zero solutions. In case it does not have a solution, we can find the 'solution' y that minimizes the norm of the error $\|Ay - b\|$. It turns out that the solution is given by $y = A^+ b$.

8.5 Problems

Problem 8.1. Find the rank r of matrix A and represent it as a product of matrices with r columns and r rows:

a) $A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$,

b) $A = \begin{pmatrix} a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \end{pmatrix}$.

Problem 8.2. Find the SVD of the matrix

a) $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$

Problem 8.3. Find eigenvalues, eigenvectors and the SVD of the matrix

a) $\begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 0 & 4 \\ 0 & 0 \end{pmatrix}$

d) $\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$

e) $\begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

Problem 8.4. Find the pseudoinverse of the matrices from 8.3. Compare this to the inverse of the matrices which are invertible.

Problem 8.5. Find the best approximation to the solution of the system of linear equations (x is a column vector of corresponding size):

a) $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} x = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}$,

b) $\begin{pmatrix} 0 & 4 \\ 0 & 0 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$,

c) $\begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix} x = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$.

9 The Fast Fourier Transform

9.1 Complex exponentials

Definition 9.1. The complex exponential is defined as

$$e^{ix} = \cos(x) + i\sin(x).$$

Why is it a natural definition? Consider the Taylor series expansion of e^{ix}:

$$e^{ix} = 1 + ix + \frac{i^2x^2}{2!} + \frac{i^3x^3}{3!} + \cdots = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \cdots.$$

Comparing this with the Taylor series expansion of $\cos(x)$ and $\sin(x)$:

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots,$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$$

we see that

$$e^{ix} = \cos(x) + i\sin(x).$$

Then we say that the Fourier series of $f(x)$ is a representation of the following form:

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}.$$

The forward Fourier transform is finding the coefficients c_k of this series. The backward Fourier transform is finding the function $f(x)$ from the coefficients c_k.

Exercise 9.1. Find the Fourier series of the function $f(x) = e^{ix}$.

Exercise 9.2. Find the Fourier series of the function $f(x) = \cos(x)$.

Exercise 9.3. Find the Fourier series of the function $f(x) = \sin(x)$.

9.2 Fourier matrix

Let us fix some natural number n and consider the set of all complex n-th roots of unity $e^{2\pi ik/n}$ for $k = 0, 1, \cdots, n-1$. If we denote $\varepsilon = e^{2\pi i/n}$ then the set of all n-th roots of unity is $\{1, \varepsilon, \varepsilon^2, \cdots, \varepsilon^{n-1}\}$. Then the Fourier matrix has the following form:

$$F_n = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \varepsilon & \varepsilon^2 & \cdots & \varepsilon^{n-1} \\ 1 & \varepsilon^2 & \varepsilon^4 & \cdots & \varepsilon^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \varepsilon^{n-1} & \varepsilon^{2(n-1)} & \cdots & \varepsilon^{(n-1)^2} \end{pmatrix}.$$

By convention, we enumerate the columns of F_n from 0 to $n-1$ and the rows from 0 to $n-1$. Then the entry $F_n(k, m)$ is $\varepsilon^{km} = e^{2\pi ikm/n}$.

Exercise 9.4. Find the Fourier matrix for $n = 4$. Is F_4 symmetric?

Exercise 9.5. Prove that $(\frac{1}{2}F_4^*)(\frac{1}{2}F_4) = I$, where F_4^* is the conjugate transpose of F_4.

A complex matrix A is called unitary if $A^*A = I$. Thus $\frac{1}{2}F_4$ is unitary, and $\overline{F}_4$ is proportional to F_4^{-1} (up to a scalar factor).

Lemma 9.2. For any natural number n we have $F_n^{-1} = \frac{1}{n}\overline{F_n}$.

Consider the following product:

$$\begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{pmatrix} = F_4 c = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \varepsilon & \varepsilon^2 & \varepsilon^3 \\ 1 & \varepsilon^2 & \varepsilon^4 & \varepsilon^6 \\ 1 & \varepsilon^3 & \varepsilon^6 & \varepsilon^9 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} c_0 + c_1 + c_2 + c_3 \\ c_0 + c_1\varepsilon + c_2\varepsilon^2 + c_3\varepsilon^3 \\ c_0 + c_1\varepsilon^2 + c_2\varepsilon^4 + c_3\varepsilon^6 \\ c_0 + c_1\varepsilon^3 + c_2\varepsilon^6 + c_3\varepsilon^9 \end{pmatrix} = \begin{pmatrix} \sum_{k=0}^{3} c_k e^{ik\cdot 0} \\ \sum_{k=0}^{3} c_k e^{ik\cdot \frac{2\pi}{4}} \\ \sum_{k=0}^{3} c_k e^{ik\cdot \frac{4\pi}{4}} \\ \sum_{k=0}^{3} c_k e^{ik\cdot \frac{6\pi}{4}} \end{pmatrix}.$$

This yields finite Fourier series for $x = 0, \frac{2\pi}{4}, \frac{4\pi}{4}, \frac{6\pi}{4}$.

How do we find c? We have $c = \frac{1}{n}\overline{F_n}\, y$.

Definition 9.3. The Discrete Fourier Transform (DFT) is the interpolation of a polynomial $p(z) = c_0 + c_1 z + \cdots + c_{n-1}z^{n-1}$ that matches n values $y_0, \ldots, y_{n-1}$ in n points $1, \ldots, \varepsilon^{n-1}$:

$$p(1) = p(\varepsilon^0) = y_0, \quad p(\varepsilon) = y_1, \quad \cdots, \quad p(\varepsilon^{n-1}) = y_{n-1}.$$

9.3 How to multiply fast?

To multiply fast, we reduce the Fourier matrix F_n to a smaller Fourier matrix $F_{n/2}$. For example, we can write F_4 as:

$$F_4 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & i \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -i \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & i^2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & i^2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For an arbitrary n we have:

$$F_{2n} = \begin{pmatrix} I_n & D_n \\ I_n & -D_n \end{pmatrix} \begin{pmatrix} F_n & 0 \\ 0 & F_n \end{pmatrix} P_{2n},$$

where I_n is the identity matrix of size n and $D_n = \operatorname{diag}[1, \varepsilon, \ldots, \varepsilon^{n-1}]$, and P_{2n} is the permutation matrix that swaps the even and odd columns of F_n.

Split y into halves y' and y'', and split c into halves $c' = (c_0, c_2, \ldots, c_{2n-2})$ and $c'' = (c_1, c_3, \ldots, c_{n-1})$. Then we have:

$$y' = F_n c', \quad y'' = F_n c''.$$

The matrix equation above shows the step from $2n$ to n as $Iy' + Dy''$ and $Iy' - Dy''$:

$$\begin{aligned} y_j &= y'_j + (\varepsilon_{2n})^j\, y''_j, \quad j = 0, \ldots, n-1 \\ y_{j+n} &= y'_j - (\varepsilon_{2n})^j\, y''_j, \quad j = 0, \ldots, n-1. \end{aligned}$$

Exercise 9.6. Write down P_8 explicitly.

Thus we can reduce from $2n$ to n by applying this step, noting that $\varepsilon_{2n}^2 = \varepsilon_n$. We can then perform this recursively.

For example, the middle matrix in F_{1024} is:

$$\begin{pmatrix} F_{512} & 0 \\ 0 & F_{512} \end{pmatrix} = \begin{pmatrix} I & D & 0 & 0 \\ I & -D & 0 & 0 \\ 0 & 0 & I & D \\ 0 & 0 & I & -D \end{pmatrix} \begin{pmatrix} F & 0 & 0 & 0 \\ 0 & F & 0 & 0 \\ 0 & 0 & F & 0 \\ 0 & 0 & 0 & F \end{pmatrix} \begin{pmatrix} \text{pick } 0, 4, 8, \ldots \\ \text{pick } 2, 6, 10, \ldots \\ \text{pick } 1, 5, 9, \ldots \\ \text{pick } 3, 7, 11, \ldots \end{pmatrix}.$$

Let us compute the complexity of this algorithm. To multiply a square $n \times n$ matrix by a vector, we need n^2 multiplications. Using the Fourier transform, we need only $\frac{1}{2} n \log_2 n$ multiplications.

Exercise 9.7. There is an amazing rule for the order in which the c's enter the FFT after all even–odd permutations. Write the numbers 0 to $n-1$ in binary (for example, $00, 01, 10, 11$ for $n = 4$). Reverse the order of those digits: $00, 10, 01, 11$. That gives the bit-reversed order $0, 2, 1, 3$ with evens before odds. The complete picture shows the c's in bit-reversed order, the $\ell = \log_2 n$ steps of the recursion, and the final output $y_0, \ldots, y_{n-1}$, which is F_n times c.

9.4 Problems

PROBLEM 9.1. Write down one step of the FFT algorithm for $n = 6$ reduced to $n = 3$.

Solution. Let $\omega = e^{2\pi i/3}$ and let $\varepsilon = e^{2\pi i/6}$, so $\omega = \varepsilon^2$.

Split the input $c = (c_0, c_1, c_2, c_3, c_4, c_5)^{\mathrm{T}}$ into evens and odds:

$$c' = (c_0, c_2, c_4)^{\mathrm{T}}, \qquad c'' = (c_1, c_3, c_5)^{\mathrm{T}}.$$

In the matrix form, one step from 6 to 3 is

$$F_6 = \begin{pmatrix} I_3 & D_3 \\ I_3 & -D_3 \end{pmatrix} \begin{pmatrix} F_3 & 0 \\ 0 & F_3 \end{pmatrix} P_6.$$

Write the matrices explicitly:

$$F_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix},$$

$$F_6 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \varepsilon & \varepsilon^2 & \varepsilon^3 & \varepsilon^4 & \varepsilon^5 \\ 1 & \varepsilon^2 & \varepsilon^4 & 1 & \varepsilon^2 & \varepsilon^4 \\ 1 & \varepsilon^3 & 1 & \varepsilon^3 & 1 & \varepsilon^3 \\ 1 & \varepsilon^4 & \varepsilon^2 & 1 & \varepsilon^4 & \varepsilon^2 \\ 1 & \varepsilon^5 & \varepsilon^4 & \varepsilon^3 & \varepsilon^2 & \varepsilon \end{pmatrix},$$

$$D_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon^2 \end{pmatrix}.$$

Since P_6 permutes the input into bit-reversed (or even–odd) order $(0, 2, 4, 1, 3, 5)$, we have:

$$P_6 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$F_6 c = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \varepsilon & \varepsilon^2 & \varepsilon^3 & \varepsilon^4 & \varepsilon^5 \\ 1 & \varepsilon^2 & \varepsilon^4 & 1 & \varepsilon^2 & \varepsilon^4 \\ 1 & \varepsilon^3 & 1 & \varepsilon^3 & 1 & \varepsilon^3 \\ 1 & \varepsilon^4 & \varepsilon^2 & 1 & \varepsilon^4 & \varepsilon^2 \\ 1 & \varepsilon^5 & \varepsilon^4 & \varepsilon^3 & \varepsilon^2 & \varepsilon \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{pmatrix}$$

and

$$\begin{pmatrix} I_3 & D_3 \\ I_3 & -D_3 \end{pmatrix} \begin{pmatrix} F_3 & 0 \\ 0 & F_3 \end{pmatrix} P_6 c = \begin{pmatrix} F_3 & D_3 F_3 \\ F_3 & -D_3 F_3 \end{pmatrix} P_6 c = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \varepsilon & \varepsilon\omega & \varepsilon\omega^2 \\ 1 & \omega^2 & \omega & \varepsilon^2 & \varepsilon^2\omega^2 & \varepsilon^2\omega \\ 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & \omega & \omega^2 & -\varepsilon & -\varepsilon\omega & -\varepsilon\omega^2 \\ 1 & \omega^2 & \omega & -\varepsilon^2 & -\varepsilon^2\omega^2 & -\varepsilon^2\omega \end{pmatrix} \begin{pmatrix} c_0 \\ c_2 \\ c_4 \\ c_1 \\ c_3 \\ c_5 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \varepsilon^2 & \varepsilon^4 & \varepsilon & \varepsilon^3 & \varepsilon^5 \\ 1 & \varepsilon^4 & \varepsilon^2 & \varepsilon^2 & 1 & \varepsilon^4 \\ 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & \varepsilon^2 & \varepsilon^4 & -\varepsilon & -\varepsilon^3 & -\varepsilon^5 \\ 1 & \varepsilon^4 & \varepsilon^2 & -\varepsilon^2 & -1 & -\varepsilon^4 \end{pmatrix} \begin{pmatrix} c_0 \\ c_2 \\ c_4 \\ c_1 \\ c_3 \\ c_5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \varepsilon^2 & \varepsilon^4 & \varepsilon & \varepsilon^3 & \varepsilon^5 \\ 1 & \varepsilon^4 & \varepsilon^2 & \varepsilon^2 & 1 & \varepsilon^4 \\ 1 & 1 & 1 & \varepsilon^3 & \varepsilon^3 & \varepsilon^3 \\ 1 & \varepsilon^2 & \varepsilon^4 & \varepsilon^4 & 1 & \varepsilon^2 \\ 1 & \varepsilon^4 & \varepsilon^2 & \varepsilon^5 & \varepsilon^3 & \varepsilon \end{pmatrix} \begin{pmatrix} c_0 \\ c_2 \\ c_4 \\ c_1 \\ c_3 \\ c_5 \end{pmatrix}.$$

Problem 9.2. Write down one step of the FFT algorithm for $n = 8$.

Problem 9.3. Write down the FFT algorithm for $n = 4$.

Problem 9.4. Let $p(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ be a polynomial of degree 3. You are given its values at the 4-th roots of unity:

$$p(1) = 8, \quad p(i) = 3 - i, \quad p(-1) = -6, \quad p(-i) = 3 + i.$$

Using the 4-point inverse DFT, recover the coefficient vector (c_0, c_1, c_2, c_3).